

DEEP REINFORCEMENT LEARNING

HOMEWORK 2 | POLICY-BASED METHODS

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1 – Policy Gradients & Variance Reduction

In value-based methods, we learned a value function and derived a deterministic policy by taking the argmax over actions. In policy-based methods, we directly parameterize the policy $\pi_\theta(a|s)$ and optimize the parameters θ to maximize the expected return $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$. Let $\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$ denote a trajectory. Assume the probability of a trajectory is given by

$$P(\tau|\theta) = P(s_0) \prod_{t=0}^T \pi_\theta(a_t|s_t) P(s_{t+1}|s_t, a_t).$$

1.1 (3 Points) The Score Function Estimator

To optimize $J(\theta)$, we need its gradient $\nabla_\theta J(\theta)$. However, we cannot simply push the gradient inside the expectation because the expectation itself depends on the parameters θ . Show that the gradient of the objective function can be rewritten as an expectation over trajectories:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t|s_t) R(\tau) \right].$$

Answer

Step 1: Notation and setup. Let a trajectory be $\tau = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$. The probability of a trajectory under policy π_θ is

$$P(\tau | \theta) = P(s_0) \prod_{t=0}^T \pi_\theta(a_t | s_t) P(s_{t+1} | s_t, a_t).$$

The expected return is

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)] = \sum_{\tau} P(\tau | \theta) R(\tau),$$

where the sum runs over all possible trajectories (or an integral in continuous spaces; the derivation is identical).

Step 2: Differentiate under the sum. We wish to compute $\nabla_\theta J(\theta)$. Under standard regularity conditions (differentiability of π_θ and a dominated convergence argument) we may interchange gradient and summation:

$$\nabla_\theta J(\theta) = \sum_{\tau} \nabla_\theta [P(\tau | \theta) R(\tau)] = \sum_{\tau} (\nabla_\theta P(\tau | \theta)) R(\tau),$$

since $R(\tau)$ does not depend on θ .

Step 3: Log-derivative trick. For any strictly positive differentiable function $f(\theta)$, the chain rule gives

$$\nabla_\theta \log f(\theta) = \frac{\nabla_\theta f(\theta)}{f(\theta)} \implies \nabla_\theta f(\theta) = f(\theta) \nabla_\theta \log f(\theta).$$

Applying this with $f(\theta) = P(\tau | \theta)$ (trajectory probabilities are non-negative; if zero for some τ the contribution is zero anyway) yields

$$\nabla_\theta P(\tau | \theta) = P(\tau | \theta) \nabla_\theta \log P(\tau | \theta).$$

Substituting into the gradient expression:

$$\nabla_{\theta} J(\theta) = \sum_{\tau} P(\tau | \theta) (\nabla_{\theta} \log P(\tau | \theta)) R(\tau).$$

Step 4: Expand the log-probability of a trajectory. Take the logarithm of the trajectory probability factorization:

$$\log P(\tau | \theta) = \log P(s_0) + \sum_{t=0}^T \log \pi_{\theta}(a_t | s_t) + \sum_{t=0}^T \log P(s_{t+1} | s_t, a_t).$$

Step 5: Gradient of the log-probability. Differentiate with respect to θ :

$$\nabla_{\theta} \log P(\tau | \theta) = \underbrace{\nabla_{\theta} \log P(s_0)}_{=0} + \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) + \underbrace{\sum_{t=0}^T \nabla_{\theta} \log P(s_{t+1} | s_t, a_t)}_{=0}.$$

The initial state distribution $P(s_0)$ and the environment transition probabilities $P(s_{t+1} | s_t, a_t)$ do not depend on the policy parameters θ ; hence their gradients are zero. We are left with

$$\nabla_{\theta} \log P(\tau | \theta) = \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t).$$

Step 6: Assemble the score function estimator. Insert the simplified gradient of $\log P(\tau | \theta)$ back into the expression from Step 3:

$$\nabla_{\theta} J(\theta) = \sum_{\tau} P(\tau | \theta) \left(\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right) R(\tau).$$

The right-hand side is exactly the expectation over trajectories drawn from π_{θ} :

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(\tau) \right].$$

This is the **score function estimator** (also known as the REINFORCE gradient). It expresses the policy gradient purely as an expectation, enabling Monte Carlo estimation from sampled trajectories.

1.2 (2 Points) The Model-Free Property

In most reinforcement learning scenarios, the environment's transition dynamics $P(s_{t+1} | s_t, a_t)$ are completely unknown to the agent. Prove why we can still compute the policy gradient $\nabla_{\theta} \log P(\tau | \theta)$ from sample trajectories without needing to model or know the environment's transition dynamics. Show exactly what happens to the $P(s_{t+1} | s_t, a_t)$ term during the derivation.

Answer

From Question 1 we have

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} [\nabla_{\theta} \log P(\tau | \theta) R(\tau)].$$

Take the log-probability of a trajectory and differentiate:

$$\begin{aligned} \log P(\tau | \theta) &= \log P(s_0) + \sum_{t=0}^T \log \pi_{\theta}(a_t | s_t) + \sum_{t=0}^T \log P(s_{t+1} | s_t, a_t), \\ \nabla_{\theta} \log P(\tau | \theta) &= \underbrace{\nabla_{\theta} \log P(s_0)}_{=0} + \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) + \underbrace{\sum_{t=0}^T \nabla_{\theta} \log P(s_{t+1} | s_t, a_t)}_{=0}. \end{aligned}$$

The initial distribution $P(s_0)$ and the transition dynamics $P(s_{t+1} | s_t, a_t)$ do **not** depend on the policy parameters θ , so their gradients vanish. Hence

$$\nabla_{\theta} \log P(\tau | \theta) = \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t).$$

Substituting back gives the REINFORCE estimator

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(\tau) \right].$$

The gradient depends **only** on the policy's score function $\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$, which is a known, computable function of the parameters θ and the observed state-action pairs. No knowledge or model of the environment's transition probabilities is ever required; the environment merely provides sample trajectories as a black-box. Thus the policy gradient is **model-free**.

1.3 (2 Points) Zero-Bias Baseline

Policy gradients suffer from high variance. To reduce this variance, we subtract a state-dependent function $b(s_t)$, known as a baseline, from our return. Prove that subtracting an arbitrary baseline $b(s_t)$ that depends only on the current state does not introduce any bias into the policy gradient estimate.

While any valid baseline $b(s_t)$ leaves the gradient unbiased, some baselines reduce variance better than others. Often, the value function $V^{\pi}(s_t)$ is used. However, it is not strictly the variance-minimizing baseline.

Answer

Setup. The (REINFORCE) policy gradient is

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R(\tau) \right].$$

We consider subtracting a state-dependent baseline $b(s_t)$, obtaining the modified estimator

$$\hat{g}_b(\tau) = \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (R(\tau) - b(s_t)).$$

We must show that the expected value of $\hat{g}_b(\tau)$ equals $\nabla_\theta J(\theta)$, i.e., the baseline introduces **no bias**.

Step 1: Decompose the expected value. Take the expectation under π_θ :

$$\mathbb{E}_\tau[\hat{g}_b(\tau)] = \mathbb{E}_\tau \left[\underbrace{\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) R(\tau)}_{=\nabla_\theta J(\theta)} \right] - \mathbb{E}_\tau \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) b(s_t) \right].$$

Thus it suffices to prove that the subtracted term is exactly zero.

Step 2: Focus on a single time step. By linearity of expectation, it is enough to show for each t :

$$\mathbb{E}_\tau[\nabla_\theta \log \pi_\theta(a_t | s_t) b(s_t)] = 0.$$

Condition on all variables up to s_t (i.e., the past). In particular $b(s_t)$ depends only on s_t , not on a_t or future states. We may write

$$\mathbb{E}_\tau[\nabla_\theta \log \pi_\theta(a_t | s_t) b(s_t)] = \mathbb{E}_{s_0, a_0, \dots, s_t} [b(s_t) \mathbb{E}_{a_t \sim \pi_\theta(\cdot | s_t)} [\nabla_\theta \log \pi_\theta(a_t | s_t)]] .$$

Step 3: The score function has zero expectation. For any fixed state s ,

$$\mathbb{E}_{a \sim \pi_\theta(\cdot | s)} [\nabla_\theta \log \pi_\theta(a | s)] = \int \pi_\theta(a | s) \frac{\nabla_\theta \pi_\theta(a | s)}{\pi_\theta(a | s)} da = \nabla_\theta \int \pi_\theta(a | s) da = \nabla_\theta 1 = 0.$$

Therefore the inner expectation in Step 2 is zero for every possible value of the conditioned variables, making the whole expectation zero.

Step 4: Conclusion. Since each time-step contribution vanishes,

$$\mathbb{E}_\tau \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) b(s_t) \right] = 0,$$

and consequently

$$\mathbb{E}_\tau \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) (R(\tau) - b(s_t)) \right] = \nabla_\theta J(\theta).$$

Subtracting **any** function $b(s_t)$ that depends only on the current state leaves the gradient estimator unbiased. This is why a baseline (e.g., the state-value function $V(s_t)$) can be freely used for variance reduction.

1.4 (3 Points) The Optimal Baseline

Derive the exact, optimal baseline $b^*(s_t)$ that strictly minimizes the variance of the policy gradient estimate.

Answer

Setup. We use the REINFORCE gradient estimator with a state-dependent baseline $b(s)$:

$$\hat{\mathbf{g}}(\tau) = \sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) (R(\tau) - b(s_t)), \quad \nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} [\hat{\mathbf{g}}(\tau)].$$

We denote $\mathbf{g}_t = \nabla_{\theta} \log \pi_{\theta}(a_t|s_t)$ and $R = R(\tau)$. The goal is to find a function $b^* : \mathcal{S} \rightarrow \mathbb{R}$ that strictly minimizes the variance

$$\text{Var}(\hat{\mathbf{g}}) = \mathbb{E}[\|\hat{\mathbf{g}}\|^2] - \|\nabla_{\theta} J(\theta)\|^2.$$

Since $\nabla_{\theta} J(\theta)$ does not depend on b , minimizing the variance is equivalent to minimizing the second moment

$$\mathcal{L}(b) = \mathbb{E}_{\tau} [\|\hat{\mathbf{g}}(\tau)\|^2].$$

Step 1: Expand the squared norm.

$$\begin{aligned} \|\hat{\mathbf{g}}\|^2 &= \left\| \sum_{t=0}^T \mathbf{g}_t (R - b(s_t)) \right\|^2 \\ &= \sum_{t=0}^T \sum_{t'=0}^T (\mathbf{g}_t^{\top} \mathbf{g}_{t'}) (R - b(s_t)) (R - b(s_{t'})). \end{aligned}$$

Let $g_{t,t'} = \mathbf{g}_t^{\top} \mathbf{g}_{t'}$. Expanding the product gives

$$(R - b(s_t))(R - b(s_{t'})) = R^2 - Rb(s_t) - Rb(s_{t'}) + b(s_t)b(s_{t'}).$$

Hence

$$\mathcal{L}(b) = \mathbb{E} \left[\sum_{t,t'} g_{t,t'} R^2 \right] - 2 \mathbb{E} \left[\sum_{t,t'} g_{t,t'} R b(s_t) \right] + \mathbb{E} \left[\sum_{t,t'} g_{t,t'} b(s_t) b(s_{t'}) \right],$$

where we used the symmetry $g_{t,t'} = g_{t',t}$ to combine the two linear terms.

Step 2: Isolate the dependence on the values $b(s)$. Introduce the indicator function $\mathbb{I}[s_t = s]$ to rewrite sums that involve $b(s_t)$ or $b(s_{t'})$.

$$\begin{aligned} \mathbb{E} \left[\sum_{t,t'} g_{t,t'} R b(s_t) \right] &= \sum_s b(s) \mathbb{E} \left[R \sum_{t,t'} g_{t,t'} \mathbb{I}[s_t = s] \right], \\ \mathbb{E} \left[\sum_{t,t'} g_{t,t'} b(s_t) b(s_{t'}) \right] &= \sum_{s,s'} b(s) b(s') \mathbb{E} \left[\sum_{t,t'} g_{t,t'} \mathbb{I}[s_t = s] \mathbb{I}[s_{t'} = s'] \right]. \end{aligned}$$

Define

$$A_{s,s'} = \mathbb{E} \left[\sum_{t,t'} g_{t,t'} \mathbb{I}[s_t = s] \mathbb{I}[s_{t'} = s'] \right], \quad c_s = \mathbb{E} \left[R \sum_{t,t'} g_{t,t'} \mathbb{I}[s_t = s] \right].$$

Then

$$\mathcal{L}(b) = \underbrace{\mathbb{E} \left[R^2 \sum_{t,t'} g_{t,t'} \right]}_{\text{independent of } b} - 2 \sum_s c_s b(s) + \sum_{s,s'} A_{s,s'} b(s) b(s').$$

Step 3: Functional derivative and first-order optimality condition. $\mathcal{L}$ is a quadratic functional in the vector $\mathbf{b} = (b(s))_{s \in \mathcal{S}}$. For each state s ,

$$\frac{\partial \mathcal{L}}{\partial b(s)} = -2c_s + 2 \sum_{s'} A_{s,s'} b(s') = 0,$$

because $\frac{\partial}{\partial b(s)} \sum_{s_1, s_2} A_{s_1, s_2} b(s_1) b(s_2) = \sum_{s'} A_{s, s'} b(s') + \sum_{s'} A_{s', s} b(s') = 2 \sum_{s'} A_{s, s'} b(s')$ (using $A_{s, s'} = A_{s', s}$, which follows from the symmetry $g_{t, t'} = g_{t', t}$ and the symmetry of the indicator product).

Thus we obtain, for every s ,

$$\sum_{s'} A_{s, s'} b(s') = c_s.$$

Step 4: Solution of the linear system. In matrix form, $\mathbf{A}\mathbf{b} = \mathbf{c}$. The matrix $\mathbf{A}$ is a Gram matrix:

$$A_{s, s'} = \mathbb{E}[\mathbf{G}_s^\top \mathbf{G}_{s'}], \quad \mathbf{G}_s = \sum_{t=0}^T \mathbf{g}_t \mathbb{I}[s_t = s].$$

$\mathbf{A}$ is symmetric and positive semidefinite. If $\mathbf{A}$ is invertible (typically true when the policy explores all states and the gradient features are linearly independent), the unique variance-minimizing baseline is

$$\mathbf{b}^* = \mathbf{A}^{-1} \mathbf{c}.$$

If $\mathbf{A}$ is singular, any minimizer satisfies the normal equations; one can use the Moore–Penrose pseudoinverse.

Step 5: Interpretation. The optimal baseline is a **gradient-weighted expectation** of the total return:

$$b^*(s) = \sum_{s'} (\mathbf{A}^{-1})_{s, s'} \mathbb{E}\left[R \sum_{t, t'} \mathbf{g}_t^\top \mathbf{g}_{t'} \mathbb{I}[s_{t'} = s']\right].$$

In the special case where we restrict the baseline to be a constant scalar $b(s) = b$ for all s , we recover the course slide's formula:

$$b_{\text{scalar}}^* = \frac{\mathbb{E}[\|\sum_t \mathbf{g}_t\|^2 R]}{\mathbb{E}[\|\sum_t \mathbf{g}_t\|^2]}.$$

The full state-dependent baseline yields strictly lower (or equal) variance by leveraging the correlation structure of the score functions across different states within a trajectory.

2 – REINFORCE

In this section, we study the Monte Carlo policy-gradient method known as REINFORCE.

The goal is to train a policy network that maximizes the expected discounted return. The core difficulty is that Monte Carlo policy-gradient estimates are unbiased but often have high variance. Therefore, this assignment compares four related estimators:

- REINFORCE without the causality trick, using the same full-episode return for every action.
- REINFORCE with the causality trick, using reward-to-go.
- REINFORCE with reward-to-go and a constant baseline.
- REINFORCE with reward-to-go and a learned value-function baseline.

2.1 (25 Points) Implementation 1: REINFORCE Implementation

Implement all four variants described above.

Answer

2.2 (3 Points)

Answer the questions inside the Jupyter notebook based on your experiments.

Answer

Algorithm 1: Vanilla REINFORCE

Input: policy π_θ , learning rate α , discount factor γ , number of episodes N

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1 for  $episode = 1, \dots, N$  do
2   Roll out one trajectory using  $a_t \sim \pi_\theta(\cdot | s_t)$ ;
3   Store rewards  $R_t$  and log-probabilities  $\log \pi_\theta(a_t | s_t)$ ;
4   Compute the full discounted return  $G_0 = \sum_{k=0}^T \gamma^k R_k$ ;
5   Form the loss  $L(\theta) = - \sum_{t=0}^T \log \pi_\theta(a_t | s_t) G_0$ ;
6   Update  $\theta$  by gradient descent on  $L(\theta)$ ;
7 end for
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3 – Policy-Induced Distributions and Gradients

A central theme in policy-based reinforcement learning is that a policy can be studied through the distributions it induces over states and actions. In finite-horizon problems, this leads to occupancy measures and flow-conservation constraints. In discounted infinite-horizon problems, similar visitation quantities appear in the gradient of the value function. Finally, in the undiscounted average-reward setting, the usual value function must be replaced by the differential value, which satisfies the Poisson equation. Together, these problems show how policy-induced visitation structure connects occupancy measures, value gradients, and average-reward analysis.

3.1 Occupancy Measures

Consider a finite-horizon Markov Decision Process (MDP) defined by the tuple (S, A, P, R, γ) with horizon H . The state space S is partitioned into H layers: $S = S_1 \cup S_2 \cup \dots \cup S_H$, where $S_1 = \{s_{\text{init}}\}$ is the initial state. For a stochastic policy $\pi : S \rightarrow \Delta(A)$ (where $\Delta(A)$ is the probability simplex over actions), the occupancy measure $q_\pi : S \times A \rightarrow [0, 1]$ is defined as the probability that the agent visits state-action pair (s, a) when starting at s_{init} and following π for H steps.

3.1.1 (2 Points) Markov Chain as a Special Case Suppose $|A| = 1$ (only one action).

- Explain how the MDP reduces to a Markov Chain (MC).
- In this MC, what does the occupancy measure q_π represent?
- If the MC were ergodic and we considered an infinite horizon ($H \rightarrow \infty$), how would the occupancy measure relate to the stationary distribution of the MC?

Answer

Step 1: Setup. We are given a finite-horizon MDP with horizon H , state space S partitioned into layers $S_1, \dots, S_H$, action space A , transition probabilities $P(s_{t+1} | s_t, a_t)$, and a policy

$\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$. The occupancy measure is defined as

$$q_\pi(s, a) = \frac{1}{H} \sum_{t=1}^H \mathbb{P}_\pi(s_t = s, a_t = a).$$

It is a probability distribution over state-action pairs.

Step 2: Reduction to a Markov Chain ($|\mathcal{A}| = 1$). If there is only a single action, say $\mathcal{A} = \{a_0\}$, then every policy π must satisfy $\pi(a_0 | s) = 1$ for all s . The action is completely determined by the state (indeed by the fact that there is no choice). Consequently the environment transition probabilities become independent of the action:

$$P(s' | s) \triangleq P(s' | s, a_0), \quad \forall s, s'.$$

The process $\langle s_0, s_1, \dots, s_H \rangle$ is now a time-homogeneous Markov chain with transition matrix $\mathbf{P}$ (where $\mathbf{P}_{s,s'} = P(s' | s)$). No decisions are made; the randomness comes only from the initial distribution $P(s_1)$ and the transition kernel $\mathbf{P}$. Thus the MDP degenerates into a Markov chain.

Step 3: Interpretation of the Occupancy Measure. With only the action a_0 , the occupancy measure simplifies to a state distribution because the action component carries no information. Define the **state occupancy measure**

$$d_\pi(s) \triangleq q_\pi(s, a_0) = \frac{1}{H} \sum_{t=1}^H \mathbb{P}(s_t = s).$$

Here $\mathbb{P}(s_t = s)$ is the probability that the Markov chain is in state s at time step t , starting from $s_1 = s_{\text{init}}$ (or according to the initial distribution). This quantity is **not** a product of transition probabilities; rather it is the expected fraction of time the chain spends in state s during the finite horizon H . In other words,

$$d_\pi(s) = \mathbb{E}_{t \sim \text{Uniform}(1, \dots, H)} [\mathbb{P}(s_t = s)],$$

the average marginal distribution of the state over the episode.

Step 4: Infinite Horizon and the Stationary Distribution. Now suppose the Markov chain is **ergodic** (irreducible and aperiodic) and we let the horizon become infinite ($H \rightarrow \infty$). By the ergodic theorem for finite-state Markov chains, there exists a unique stationary distribution $\mu = (\mu(s))_{s \in \mathcal{S}}$ satisfying

$$\mathbf{P}\mu = \mu, \quad \sum_s \mu(s) = 1, \quad \mu(s) > 0.$$

Moreover, for any initial distribution, the time-averaged fraction of visits to a state converges almost surely to the stationary probability:

$$\lim_{H \rightarrow \infty} \frac{1}{H} \sum_{t=1}^H \mathbf{1}\{s_t = s\} = \mu(s) \quad \text{a.s.},$$

and consequently the expected fraction converges as well:

$$\lim_{H \rightarrow \infty} \frac{1}{H} \sum_{t=1}^H \mathbb{P}(s_t = s) = \mu(s).$$

Thus, as the horizon grows unbounded, the state occupancy measure $d_\pi(s)$ tends to the stationary distribution of the Markov chain:

$$\lim_{H \rightarrow \infty} d_\pi(s) = \mu(s).$$

In summary:

- With a single action the MDP becomes a Markov chain.
- The occupancy measure reduces to the expected time-averaged state distribution.
- For an ergodic chain, this time-averaged distribution converges to the stationary distribution as the horizon goes to infinity.

3.1.2 (2 Points) Properties of the Occupancy Measure Show that for any policy π , the induced occupancy measure q_π satisfies the following linear constraints:

$$\sum_{a \in A} q_\pi(s_{\text{init}}, a) = 1, \quad (1)$$

and for each $h = 1, \dots, H - 1$ and each $s' \in S_{h+1}$,

$$\sum_{s \in S_h} \sum_{a \in A} q_\pi(s, a) P(s' | s, a) = \sum_{a \in A} q_\pi(s', a). \quad (2)$$

(Interpretation: The second equation is a flow conservation condition: the probability of entering state s' equals the probability of being in s' .)

Answer

Step 1: Definition of the occupancy measure. The MDP horizon is H , and the state space is partitioned by time step: $\mathcal{S} = \mathcal{S}_1 \cup \dots \cup \mathcal{S}_H$, with $\mathcal{S}_1 = \{s_{\text{init}}\}$. For a policy π , the occupancy measure $q_\pi(s, a)$ is the probability that the agent is in state s and takes action a at the time step corresponding to s 's layer. Formally, if $s \in \mathcal{S}_h$, then

$$q_\pi(s, a) = \mathbb{P}_\pi(s_h = s, a_h = a).$$

Because the agent is exactly in one state-action pair at each time step, for each layer h we have

$$\sum_{s \in \mathcal{S}_h} \sum_{a \in A} q_\pi(s, a) = 1.$$

Step 2: Initial-state constraint. At time $h = 1$, the agent always starts in s_{init} . Therefore

$$\sum_{a \in A} q_\pi(s_{\text{init}}, a) = \sum_{a \in A} \mathbb{P}_\pi(s_1 = s_{\text{init}}, a_1 = a) = \mathbb{P}_\pi(s_1 = s_{\text{init}}) = 1.$$

Thus the first constraint holds:

$$\sum_{a \in A} q_\pi(s_{\text{init}}, a) = 1.$$

Step 3: Flow-conservation constraint. Let $h \in \{1, \dots, H - 1\}$ and consider a state $s' \in \mathcal{S}_{h+1}$. The probability of arriving at s' at time $h + 1$ is obtained by marginalizing over all

possible previous state–action pairs at time h :

$$\mathbb{P}_\pi(s_{h+1} = s') = \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} \mathbb{P}_\pi(s_h = s, a_h = a) P(s' | s, a).$$

Recognising $\mathbb{P}_\pi(s_h = s, a_h = a) = q_\pi(s, a)$, we have

$$\mathbb{P}_\pi(s_{h+1} = s') = \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q_\pi(s, a) P(s' | s, a). \quad (1)$$

On the other hand, the probability that the agent is in s' at time $h + 1$ and performs some action is

$$\mathbb{P}_\pi(s_{h+1} = s') = \sum_{a \in \mathcal{A}} \mathbb{P}_\pi(s_{h+1} = s', a_{h+1} = a) = \sum_{a \in \mathcal{A}} q_\pi(s', a). \quad (2)$$

Equating (1) and (2) yields the flow-conservation condition for every $h = 1, \dots, H - 1$ and every $s' \in \mathcal{S}_{h+1}$:

$$\boxed{\sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q_\pi(s, a) P(s' | s, a) = \sum_{a \in \mathcal{A}} q_\pi(s', a)}.$$

Interpretation. The left-hand side is the total probability mass entering s' at time $h + 1$; the right-hand side is the total probability mass occupying s' at that same time step. Their equality is a **flow conservation law**: in the deterministic layered timeline, the amount of probability flowing into a state equals the amount of probability staying in (and then leaving from) that state. These linear constraints characterize exactly which distributions over state–action pairs can be realized by some policy π .

3.1.3 (4 Points) From Constraints to Occupancy Measure Now suppose we have a function $q : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ that satisfies constraints (1) and (2) above. Define a policy π by

$$\pi(a | s) = \frac{q(s, a)}{\sum_{a' \in \mathcal{A}} q(s, a')} \quad \text{if } \sum_{a'} q(s, a') > 0,$$

and let $\pi(\cdot | s)$ be an arbitrary distribution over $\mathcal{A}$ otherwise. Prove that q is the occupancy measure induced by π , i.e., $q = q_\pi$.

Answer

Step 1: Setup and notation. We are given a function $q : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ that satisfies the constraints (1) and (2) from the previous subquestion:

1. $\sum_{a \in \mathcal{A}} q(s_{\text{init}}, a) = 1$.
2. For every $h = 1, \dots, H - 1$ and every $s' \in \mathcal{S}_{h+1}$,

$$\sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q(s, a) P(s' | s, a) = \sum_{a \in \mathcal{A}} q(s', a).$$

Define a policy π by

$$\pi(a | s) = \frac{q(s, a)}{\sum_{a' \in \mathcal{A}} q(s, a')} \quad \text{if } \sum_{a'} q(s, a') > 0,$$

and let $\pi(\cdot | s)$ be an arbitrary distribution over $\mathcal{A}$ otherwise. We must prove that q is exactly the occupancy measure induced by π , i.e., for every h and every $s \in \mathcal{S}_h$, $a \in \mathcal{A}$,

$$q(s, a) = \mathbb{P}_\pi(s_h = s, a_h = a) \equiv q_\pi(s, a).$$

Step 2: Base case ($h = 1$). For $h = 1$, the only state is $s_{\text{init}} \in \mathcal{S}_1$. By constraint (1),

$$\sum_{a \in \mathcal{A}} q(s_{\text{init}}, a) = 1.$$

Hence the denominator in the definition of π for s_{init} equals 1, and we obtain

$$\pi(a | s_{\text{init}}) = q(s_{\text{init}}, a) \quad \forall a.$$

Under policy π , the probability of starting in s_{init} is 1, so

$$\mathbb{P}_\pi(s_1 = s_{\text{init}}, a_1 = a) = 1 \cdot \pi(a | s_{\text{init}}) = q(s_{\text{init}}, a).$$

Thus $q(s, a) = q_\pi(s, a)$ for all $s \in \mathcal{S}_1$, $a \in \mathcal{A}$.

Step 3: Inductive hypothesis. Assume that for some $h \in \{1, \dots, H-1\}$,

$$q(s, a) = \mathbb{P}_\pi(s_h = s, a_h = a) \quad \forall s \in \mathcal{S}_h, a \in \mathcal{A}.$$

Step 4: Inductive step for state probabilities. We first compute the marginal probability of reaching a state $s' \in \mathcal{S}_{h+1}$ at time $h+1$:

$$\begin{aligned} \mathbb{P}_\pi(s_{h+1} = s') &= \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} \mathbb{P}_\pi(s_h = s, a_h = a) P(s' | s, a) \\ &= \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q(s, a) P(s' | s, a) \quad (\text{by the inductive hypothesis}). \end{aligned}$$

Applying constraint (2) exactly matches the right-hand side to $\sum_{a \in \mathcal{A}} q(s', a)$. Hence

$$\mathbb{P}_\pi(s_{h+1} = s') = \sum_{a \in \mathcal{A}} q(s', a). \quad (3)$$

Step 5: Inductive step for state-action probabilities. Now for any $a' \in \mathcal{A}$,

$$\mathbb{P}_\pi(s_{h+1} = s', a_{h+1} = a') = \mathbb{P}_\pi(s_{h+1} = s') \pi(a' | s').$$

We distinguish two cases:

Case 1: $\sum_a q(s', a) > 0$. Then by definition $\pi(a' | s') = \frac{q(s', a')}{\sum_a q(s', a)}$, and using (3),

$$\mathbb{P}_\pi(s_{h+1} = s', a_{h+1} = a') = \left(\sum_a q(s', a) \right) \cdot \frac{q(s', a')}{\sum_a q(s', a)} = q(s', a').$$

Case 2: $\sum_a q(s', a) = 0$. Then (3) gives $\mathbb{P}_\pi(s_{h+1} = s') = 0$, so the left-hand side is 0. Moreover, since all $q(\cdot, \cdot)$ are non-negative, $\sum_a q(s', a) = 0$ forces $q(s', a') = 0$. Thus $0 = 0$, and equality holds regardless of the arbitrary choice of $\pi(\cdot | s')$.

In both cases, $q(s', a') = \mathbb{P}_\pi(s_{h+1} = s', a_{h+1} = a')$ for every $s' \in \mathcal{S}_{h+1}$, $a' \in \mathcal{A}$.

Step 6: Conclusion. By induction on $h = 1, \dots, H$, we have shown

$$q(s, a) = \mathbb{P}_\pi(s_h = s, a_h = a) = q_\pi(s, a)$$

for all $s \in \mathcal{S}_h$, $a \in \mathcal{A}$. Therefore the given q is exactly the occupancy measure induced by the policy π defined from q . The constraints (1) and (2) are not only necessary but also **sufficient** for a function q to be a valid occupancy measure.

3.1.4 (1 Point) Utility of the Occupancy Measure Formulation

- Express the expected total loss $v_\pi(s_{\text{init}})$ in terms of q_π and γ .
- Discuss one advantage and one disadvantage of optimizing over the space of occupancy measures (subject to constraints (1) and (2)) compared to optimizing directly over the policy space. Hint for advantage: Consider the structure of the objective and constraints. Hint for disadvantage: Think about the dimensionality and the relationship between q and π .

Answer

Part (a): Expressing the expected total loss.

Given the layered MDP with horizon H and a discount factor $\gamma \in [0, 1]$, the expected total loss (or return) starting from s_{init} is

$$v_\pi(s_{\text{init}}) = \mathbb{E}_\pi \left[\sum_{t=1}^H \gamma^{t-1} r(s_t, a_t) \right],$$

where $r(s, a)$ is the instantaneous loss incurred in state s upon taking action a . Using the definition of the occupancy measure $q_\pi(s, a) = \mathbb{P}_\pi(s_h = s, a_h = a)$ for $s \in \mathcal{S}_h$, we can rewrite the expectation step by step:

$$v_\pi(s_{\text{init}}) = \sum_{h=1}^H \gamma^{h-1} \sum_{s \in \mathcal{S}_h} \sum_{a \in \mathcal{A}} q_\pi(s, a) r(s, a).$$

In compact form,

$$v_\pi(s_{\text{init}}) = \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \gamma^{\text{layer}(s)-1} q_\pi(s, a) r(s, a),$$

where $\text{layer}(s)$ denotes the time step h such that $s \in \mathcal{S}_h$.

Part (b): Advantage and disadvantage of optimizing over q versus over π .

- Advantage – Linear programming formulation.** The objective $v_\pi(s_{\text{init}})$ is **linear** in the variables $q(s, a)$ (after incorporating the discount weights), and the constraints (1) and (2) are also linear. Therefore the problem of finding an optimal policy can be cast as a **linear program**:

$$\max_q \sum_{s,a} \gamma^{\text{layer}(s)-1} q(s, a) r(s, a) \quad \text{s.t.} \quad (1), (2) \text{ and } q(s, a) \geq 0.$$

Linear programs are convex and can be solved to global optimality using efficient algorithms (e.g., the simplex method or interior-point methods). There are no issues with local optima or learning rates, unlike policy gradient methods.

- Disadvantage – Curse of dimensionality.** The optimization variable q lives in $\mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$, which can be extremely large in practical problems. The constraints (2) also scale with the number of states and layers. Moreover, the policy must be recovered from q via $\pi(a|s) \propto q(s, a)$, which only makes sense in tabular settings. This direct approach does **not** scale to high-dimensional or continuous state spaces where function approximation (e.g., neural networks) is necessary. Policy gradient methods, by contrast, can easily incorporate function approximation and sample-based estimates.

3.2 Discounted Visitation and the Gradient of v_π

In a discounted MDP, it holds for any $s \in \mathcal{S}$ that

$$\nabla_\theta v_\pi(s) = \sum_{s' \in \mathcal{S}} \Pr(s' | s) \sum_{a \in \mathcal{A}} \nabla_\theta \pi(a | s', \theta) q_\pi(s', a),$$

where

$$\Pr(s' | s) := [(I_n - \gamma P_\pi)^{-1}]_{ss'}$$

is the discounted total probability of transitioning from s to s' under policy π . Here, $[\cdot]_{ss'}$ denotes the entry in the s -th row and s' -th column.

3.2.1 (5 Points) The Gradient of v_π Prove that the above equation holds for all $s \in \mathcal{S}$ in a discounted MDP.

Answer

Step 1: Notation and setup. We consider an infinite-horizon discounted MDP with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P(s' | s, a)$, discount factor $\gamma \in [0, 1)$, and a parameterised policy $\pi_\theta(a | s)$. For any policy π , let

- $v_\pi(s)$ be the state-value function: $v_\pi(s) = \mathbb{E}_\pi [\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s]$.
- $q_\pi(s, a)$ be the action-value function: $q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) v_\pi(s')$. (The problem statement uses q_π to denote the action-value; this is standard notation in the policy gradient literature, not to be confused with the occupancy measure of the previous section.)
- The discounted total transition probability from s to s' under π is $\Pr(s' | s) = [(I - \gamma P_\pi)^{-1}]_{ss'}$, where P_π is the state-to-state transition matrix under π :

$$P_\pi(s' | s) = \sum_{a \in \mathcal{A}} \pi(a | s) P(s' | s, a).$$

We must prove

$$\nabla_\theta v_\pi(s) = \sum_{s' \in \mathcal{S}} \Pr(s' | s) \sum_{a \in \mathcal{A}} \nabla_\theta \pi(a | s', \theta) q_\pi(s', a). \quad (3)$$

Step 2: Differentiate the Bellman equation for v_π . The state-value function satisfies

$$v_\pi(s) = \sum_a \pi(a | s) q_\pi(s, a). \quad (1)$$

Take the gradient with respect to θ :

$$\nabla_\theta v_\pi(s) = \underbrace{\sum_a \nabla_\theta \pi(a | s) q_\pi(s, a)}_{\text{term A}} + \underbrace{\sum_a \pi(a | s) \nabla_\theta q_\pi(s, a)}_{\text{term B}}. \quad (2)$$

Step 3: Express $\nabla_\theta q_\pi(s, a)$. From the definition $q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) v_\pi(s')$, note that $r(s, a)$ and $P(s' | s, a)$ do not depend on θ . Hence

$$\nabla_\theta q_\pi(s, a) = \gamma \sum_{s'} P(s' | s, a) \nabla_\theta v_\pi(s'). \quad (3)$$

Insert (3) into term B of (2):

$$\begin{aligned} \text{term B} &= \sum_a \pi(a | s) \left(\gamma \sum_{s'} P(s' | s, a) \nabla_{\theta} v_{\pi}(s') \right) \\ &= \gamma \sum_{s'} \left(\sum_a \pi(a | s) P(s' | s, a) \right) \nabla_{\theta} v_{\pi}(s'). \end{aligned}$$

The inner sum is exactly $P_{\pi}(s' | s)$. So

$$\text{term B} = \gamma \sum_{s'} P_{\pi}(s' | s) \nabla_{\theta} v_{\pi}(s'). \quad (4)$$

Step 4: Recursive relation for $\nabla_{\theta} v_{\pi}$. Substituting (4) back into (2) yields, for every s ,

$$\nabla_{\theta} v_{\pi}(s) = \sum_a \nabla_{\theta} \pi(a | s) q_{\pi}(s, a) + \gamma \sum_{s'} P_{\pi}(s' | s) \nabla_{\theta} v_{\pi}(s'). \quad (5)$$

Write this in vector form. Define

- $\mathbf{g} \in \mathbb{R}^{|S|}$ with components $g(s) = \nabla_{\theta} v_{\pi}(s)$ (each component is a vector of dimension $|\theta|$, but we treat them as vectors in a suitable space and operations are element-wise).
- $\mathbf{b} \in \mathbb{R}^{|S|}$ with $b(s) = \sum_a \nabla_{\theta} \pi(a | s) q_{\pi}(s, a)$.

Then (5) becomes the linear system

$$\mathbf{g} = \mathbf{b} + \gamma P_{\pi} \mathbf{g}. \quad (6)$$

Step 5: Solve the linear system. Rewrite (6) as $(I - \gamma P_{\pi}) \mathbf{g} = \mathbf{b}$. Since $\gamma \in [0, 1)$ and P_{π} is a stochastic matrix, $I - \gamma P_{\pi}$ is invertible and

$$\mathbf{g} = (I - \gamma P_{\pi})^{-1} \mathbf{b}.$$

Expanding the inverse in a Neumann series gives $(I - \gamma P_{\pi})^{-1} = \sum_{k=0}^{\infty} \gamma^k P_{\pi}^k$, which matches the interpretation of discounted total transition probabilities. Component-wise, for each s ,

$$\nabla_{\theta} v_{\pi}(s) = \sum_{s'} [(I - \gamma P_{\pi})^{-1}]_{ss'} b(s') = \sum_{s'} \Pr(s' | s) \sum_a \nabla_{\theta} \pi(a | s') q_{\pi}(s', a).$$

Step 6: Conclusion. We have derived exactly Equation (3). The discounted visitation probabilities $\Pr(s' | s)$ naturally arise from the Neumann series of the policy transition matrix, and the action-value function $q_{\pi}(s', a)$ appears as the coefficient for the policy gradient. This is the fundamental **policy gradient theorem** for discounted MDPs.

3.3 Undiscounted Gradients

In the undiscounted case, it is necessary to redefine state and action values. Since the undiscounted sum of the rewards, $\mathbb{E}[R_{t+1} + R_{t+2} + \dots | S_t = s]$, may diverge, the state and action values are defined in a special way:

$$v_{\pi}(s) \doteq \mathbb{E}[(R_{t+1} - \bar{r}_{\pi}) + (R_{t+2} - \bar{r}_{\pi}) + \dots | S_t = s],$$

$$q_\pi(s, a) \doteq \mathbb{E}[(R_{t+1} - \bar{r}_\pi) + (R_{t+2} - \bar{r}_\pi) + \dots | S_t = s, A_t = a],$$

where $\bar{r}_\pi$ is the average reward, which is determined when π is given. There are different names for $v_\pi(s)$ in the literature such as the differential reward or bias. Note that in this part $q_\pi(s, a)$ denotes the action-value function, following standard reinforcement learning notation. It can be verified that the state value defined above satisfies the following Bellman-like equation:

$$v_\pi(s) = \sum_a \pi(a | s, \theta) \left[\sum_r p(r | s, a)(r - \bar{r}_\pi) + \sum_{s'} p(s' | s, a)v_\pi(s') \right].$$

Since $v_\pi(s) = \sum_{a \in \mathcal{A}} \pi(a | s, \theta) q_\pi(s, a)$, it holds that $q_\pi(s, a) = \sum_r p(r | s, a)(r - \bar{r}_\pi) + \sum_{s'} p(s' | s, a)v_\pi(s')$. The matrix-vector form of this equation is

$$v_\pi = r_\pi - \bar{r}_\pi \mathbf{1}_n + P_\pi v_\pi,$$

where $\mathbf{1}_n = [1, \dots, 1]^\top \in \mathbb{R}^n$. This equation is known as the Poisson equation.

3.3.1 (5 Points) Solution of the Poisson Equation

Show that

$$v_\pi^* = (I_n - P_\pi + \mathbf{1}_n d_\pi^\top)^{-1} r_\pi$$

is a solution of the Poisson equation. Here, the distribution d_π is selected such that it is stationary distribution under π , satisfying

$$d_\pi^\top P_\pi = d_\pi^\top,$$

where P_π is the state transition probability matrix.

Answer

Step 1: Setup. We work in an undiscounted infinite-horizon MDP with a given policy π .

Let

- $r_\pi \in \mathbb{R}^n$ be the vector of expected immediate rewards: $r_\pi(s) = \sum_a \pi(a | s) \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$.
- P_π be the state-transition matrix under π : $P_\pi(s' | s) = \sum_a \pi(a | s) P(s' | s, a)$.
- d_π be the stationary distribution over states under π , satisfying $d_\pi^\top P_\pi = d_\pi^\top$, $d_\pi^\top \mathbf{1}_n = 1$.
- $\bar{r}_\pi = d_\pi^\top r_\pi$ is the average reward per time step.

The Poisson equation (differential value equation) for the bias v_π is

$$v_\pi = r_\pi - \bar{r}_\pi \mathbf{1}_n + P_\pi v_\pi. \quad (6)$$

We must show that

$$v_\pi^* = (I_n - P_\pi + \mathbf{1}_n d_\pi^\top)^{-1} r_\pi \quad (7)$$

is a solution of (6).

Step 2: Define the invertible matrix. Set $M = I_n - P_\pi + \mathbf{1}_n d_\pi^\top$. The problem assumes M is invertible (this can be shown using properties of ergodic Markov chains). Then $v_\pi^* = M^{-1} r_\pi$.

Step 3: Express $I_n - P_\pi$ in terms of M . From the definition,

$$M = (I_n - P_\pi) + \mathbf{1}_n d_\pi^\top \implies I_n - P_\pi = M - \mathbf{1}_n d_\pi^\top.$$

Step 4: Compute $(I_n - P_\pi)v_\pi^*$. Multiply both sides of $v_\pi^* = M^{-1}r_\pi$ by $I_n - P_\pi$:

$$\begin{aligned}(I_n - P_\pi)v_\pi^* &= (M - \mathbf{1}_n d_\pi^\top)M^{-1}r_\pi \\ &= MM^{-1}r_\pi - \mathbf{1}_n d_\pi^\top M^{-1}r_\pi \\ &= r_\pi - \mathbf{1}_n (d_\pi^\top v_\pi^*).\end{aligned}$$

Step 5: Evaluate $d_\pi^\top v_\pi^*$. We first compute $d_\pi^\top M$:

$$d_\pi^\top M = d_\pi^\top (I_n - P_\pi + \mathbf{1}_n d_\pi^\top) = \underbrace{d_\pi^\top - d_\pi^\top P_\pi}_{=0} + d_\pi^\top \mathbf{1}_n d_\pi^\top.$$

Since $d_\pi^\top \mathbf{1}_n = 1$, we obtain $d_\pi^\top M = d_\pi^\top$. Multiplying on the right by M^{-1} gives

$$d_\pi^\top = d_\pi^\top M^{-1}.$$

Hence

$$d_\pi^\top v_\pi^* = d_\pi^\top (M^{-1}r_\pi) = (d_\pi^\top M^{-1})r_\pi = d_\pi^\top r_\pi = \bar{r}_\pi. \quad (\text{B})$$

Step 6: Substitute back into the Poisson equation. Insert result (B) into equation (A):

$$(I_n - P_\pi)v_\pi^* = r_\pi - \bar{r}_\pi \mathbf{1}_n.$$

Rearranging,

$$v_\pi^* = r_\pi - \bar{r}_\pi \mathbf{1}_n + P_\pi v_\pi^*,$$

which is exactly the Poisson equation (6).

Step 7: Conclusion. We have shown that v_π^* defined in (7) satisfies the Poisson equation for all states. The matrix $I_n - P_\pi + \mathbf{1}_n d_\pi^\top$ resolves the singularity of $I_n - P_\pi$ and selects a particular bias v_π^* that is consistent with the average reward $\bar{r}_\pi = d_\pi^\top r_\pi$.

3.3.2 (6 Points) General Solution of the Poisson Equation Show that any solution of the Poisson equation has the following form:

$$v^\pi = v_\pi^* + c \mathbf{1}_n,$$

where $c \in \mathbb{R}$.

Answer

Step 1: Setup. We work in the same undiscounted setting as the previous subquestion.

The Poisson equation for a given policy π is

$$v = r - \bar{r} \mathbf{1}_n + P v, \quad (6)$$

where $r = r_\pi$, $\bar{r} = \bar{r}_\pi$, $P = P_\pi$, and $\mathbf{1}_n = (1, \dots, 1)^\top$. From the previous subquestion we have one particular solution

$$v^* = (I_n - P + \mathbf{1}_n d^\top)^{-1} r,$$

with $d^\top P = d^\top$. We need to prove that **any** solution v of (6) has the form

$$v = v^* + c \mathbf{1}_n, \quad c \in \mathbb{R}. \quad (8)$$

Step 2: Subtract the particular solution. Suppose v satisfies (6). Since v^* also satisfies (6) (proved in the previous subquestion), subtract the two equations:

$$\begin{aligned} v &= r - \bar{r} \mathbf{1}_n + Pv, \\ v^* &= r - \bar{r} \mathbf{1}_n + Pv^*. \end{aligned}$$

Subtracting gives

$$v - v^* = P(v - v^*).$$

Define $\Delta = v - v^*$. Then

$$P\Delta = \Delta. \tag{9}$$

Step 3: Solve the homogeneous equation $P\Delta = \Delta$. Equation (9) says that Δ is a right eigenvector of the stochastic matrix P with eigenvalue 1. We now use a basic property of Markov chains:

If the Markov chain with transition matrix P is irreducible (or more generally, if the only solutions to $Px = x$ are the constant vectors), then Δ must be proportional to $\mathbf{1}_n$.

Justification: For a stochastic matrix, $\mathbf{1}_n$ is always a right eigenvector with eigenvalue 1 because $P\mathbf{1}_n = \mathbf{1}_n$. When the chain is irreducible, the Perron–Frobenius theorem guarantees that the eigenspace associated with the dominant eigenvalue 1 is one-dimensional, so any vector satisfying $Px = x$ is a scalar multiple of $\mathbf{1}_n$. In the average-reward MDP setting, this irreducibility (or unichain) assumption is standard to ensure well-posedness of the differential value function.

Therefore $\Delta = c \mathbf{1}_n$ for some constant $c \in \mathbb{R}$.

Step 4: Conclude. Since $v - v^* = \Delta = c \mathbf{1}_n$, we have

$$v = v^* + c \mathbf{1}_n.$$

Thus every solution of the Poisson equation (6) is obtained by taking the particular solution v^* and adding an arbitrary constant vector. The constant c corresponds to the fact that the bias function v is defined only up to an additive constant; this degree of freedom reflects the absolute level of the differential rewards.

4 – Natural Policy Gradient (NPG)

The usual policy gradient tells us how to improve the objective by changing the policy parameters. But the parameters themselves are not the real object of interest. The real object is the policy distribution. A policy parameter vector is only a coordinate system for a family of probability distributions.

This distinction matters. If we change coordinates, the same policy family may look different in parameter space, and an ordinary Euclidean gradient step may behave differently. The natural policy gradient corrects this by measuring the size of a policy update using the geometry of probability distributions.

By the end of the section, you should see why the natural policy gradient can be interpreted as a direction that moves the policy toward the action choices suggested by a greedy policy-improvement step.

4.1 Average-Reward Policy Optimization

We consider a finite Markov decision process with state space S , action space A , reward function R , and transition dynamics P .

The agent uses a stochastic policy $\pi_\theta(a|s)$, where $\theta \in \mathbb{R}^m$ parameterizes the policy. Unlike the discounted setting, here we consider a continuing task. The performance objective is not

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right].$$

Instead, the objective is the long-run average reward per time step. Intuitively, if a policy is run for a very long time, we ask: "How much reward does it collect per step on average?"

Under an ergodicity assumption (stable long-run distribution), each policy π induces a stationary state distribution $\rho^\pi(s)$. The average-reward objective is

$$\eta(\pi) = \sum_{s \in S} \rho^\pi(s) \sum_{a \in A} \pi(a|s) R(s, a).$$

Equivalently, one may view it as the infinite-horizon time average

$$\eta(\pi) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R(s_t, a_t) \right].$$

The optimization problem is therefore

$$\max_{\theta \in \mathbb{R}^m} \eta(\theta).$$

4.2 The Differential Action-Value Function

In the discounted setting, the action-value function measures the expected discounted future reward after taking action a in state s .

In the average-reward setting, we instead measure future reward relative to the policy's long-run average reward. This gives the differential action-value function:

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s, a_0 = a \right].$$

This quantity tells if we start from (s, a) , how much better or worse the future is compared to the average reward rate.

For the next task, assume that S and A are finite, rewards are bounded, and under the fixed policy π , the induced Markov chain is ergodic.

4.2.1 (4 Points) Finiteness of the Differential Action-Value Function Prove that, under these assumptions,

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s, a_0 = a \right]$$

is finite for every state-action pair (s, a) .

Answer

Assumptions. The state space $\mathcal{S}$ and action space $\mathcal{A}$ are finite. Rewards are bounded: there exists $R_{\max}$ such that $|R(s, a)| \leq R_{\max}$ for all (s, a) . The policy π is fixed and induces an **ergodic** Markov chain on $\mathcal{S}$ (irreducible and aperiodic).

Step 1: The average reward $\eta(\pi)$ is finite. Under ergodicity, the state process has a unique stationary distribution $\rho^\pi(s) > 0$ with $\sum_s \rho^\pi(s) = 1$ and $\rho^\pi = \rho^\pi P^\pi$. The long-run average

reward per step is

$$\eta(\pi) = \sum_{s \in \mathcal{S}} \rho^\pi(s) \sum_{a \in \mathcal{A}} \pi(a|s) R(s, a).$$

Because rewards are bounded, $|\sum_a \pi(a|s) R(s, a)| \leq R_{\max}$, and therefore

$$|\eta(\pi)| \leq \sum_s \rho^\pi(s) R_{\max} = R_{\max} < \infty.$$

So $\eta(\pi)$ is a finite constant.

Step 2: The differential state-value function and its Poisson equation. Define the differential state-value function (bias)

$$v^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s \right].$$

We show that v^π satisfies the Poisson equation

$$v^\pi = r^\pi - \eta(\pi) \mathbf{1}_n + P^\pi v^\pi,$$

where $r^\pi(s) = \sum_a \pi(a|s) R(s, a)$ and $P^\pi(s'|s) = \sum_a \pi(a|s) P(s'|s, a)$.

Derivation: Separate the first time step from the rest of the infinite sum:

$$\begin{aligned} v^\pi(s) &= \mathbb{E}_\pi \left[(R(s_0, a_0) - \eta(\pi)) + \sum_{t=1}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s \right] \\ &= \mathbb{E}_\pi [R(s_0, a_0) - \eta(\pi) \mid s_0 = s] + \mathbb{E}_\pi \left[\sum_{t=1}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s \right]. \end{aligned}$$

The first term is simply $r^\pi(s) - \eta(\pi)$ because $a_0 \sim \pi(\cdot|s)$. For the second term, condition on the first action and next state:

$$\begin{aligned} &\mathbb{E}_\pi \left[\sum_{t=1}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s \right] \\ &= \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) \mathbb{E}_\pi \left[\sum_{t=1}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_1 = s' \right]. \end{aligned}$$

By the time-homogeneity of the MDP and the Markov property, the inner expectation is precisely $v^\pi(s')$ (the process after time 1 is a new trajectory with initial state s'). Therefore

$$v^\pi(s) = r^\pi(s) - \eta(\pi) + \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) v^\pi(s').$$

The double sum is exactly $\sum_{s'} P^\pi(s'|s) v^\pi(s')$. Collecting for all states gives the vector equation

$$\boxed{v^\pi = r^\pi - \eta(\pi) \mathbf{1}_n + P^\pi v^\pi}. \quad (1)$$

This is the **Poisson equation** for the average-reward MDP.

Step 3: The bias v^π is finite. Equation (1) is a linear system. Its solution structure was studied in Section 3.3. From the solution of the Poisson equation (Question 6 of that section), a particular solution is

$$v^* = (I - P^\pi + \mathbf{1}_n(\rho^\pi)^\top)^{-1} r^\pi. \quad (2)$$

We must verify that the matrix $M = I - P^\pi + \mathbf{1}_n(\rho^\pi)^\top$ is invertible. Proof of invertibility:

- Since ρ^π is the stationary distribution, $\rho^\pi(I - P^\pi) = 0$.
- Suppose $Mx = 0$. Multiply on the left by ρ^π :

$$0 = \rho^\pi(I - P^\pi)x + \rho^\pi \mathbf{1}_n (\rho^\pi)^\top x = 0 + (\rho^\pi \mathbf{1}_n)(\rho^\pi x) = 1 \cdot (\rho^\pi x),$$

so $\rho^\pi x = 0$.

- Then $0 = (I - P^\pi)x + \mathbf{1}_n(\rho^\pi x) = (I - P^\pi)x$, hence x is in the nullspace of $I - P^\pi$. For an ergodic chain, $\ker(I - P^\pi) = \text{span}\{\mathbf{1}_n\}$, so $x = c\mathbf{1}_n$.
- But $\rho^\pi x = c\rho^\pi \mathbf{1}_n = c = 0$, therefore $x = 0$. Thus M is injective and, being square, invertible.

Because $\mathcal{S}$ is finite, M^{-1} is a finite matrix (all its entries are real numbers). Each entry of r^π satisfies $|r^\pi(s)| \leq R_{\max}$, so every component of v^* in (2) is a finite real number. The general solution of the Poisson equation is $v^* + c\mathbf{1}_n$ (from the later subquestion). Adding a constant preserves finiteness. Hence **every** bias function v^π that satisfies (1) – including the specific v^π defined as the expected differential sum – is finite for all states.

Step 4: Express $Q^\pi(s, a)$ in terms of v^π . The differential action-value function is

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s, a_0 = a \right].$$

Again split the sum into $t = 0$ and $t \geq 1$:

$$\begin{aligned} Q^\pi(s, a) &= (R(s, a) - \eta(\pi)) + \mathbb{E}_\pi \left[\sum_{t=1}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_0 = s, a_0 = a \right] \\ &= R(s, a) - \eta(\pi) + \sum_{s'} P(s'|s, a) \mathbb{E}_\pi \left[\sum_{t=1}^{\infty} (R(s_t, a_t) - \eta(\pi)) \mid s_1 = s' \right] \\ &= R(s, a) - \eta(\pi) + \sum_{s'} P(s'|s, a) v^\pi(s'). \end{aligned} \tag{3}$$

Step 5: $Q^\pi(s, a)$ is finite. We examine each term in (3):

- $R(s, a)$ is bounded, hence finite.
- $\eta(\pi)$ is finite (Step 1).
- $v^\pi(s')$ is finite for every s' (Step 3).
- The sum $\sum_{s'} P(s'|s, a) v^\pi(s')$ is a convex combination of finitely many finite numbers, so it is finite.

Therefore $Q^\pi(s, a)$ is a finite real number for every state-action pair (s, a) .

Conclusion. Under the assumptions of finite state/action spaces, bounded rewards, and an ergodic induced Markov chain, the differential action-value function $Q^\pi(s, a)$ exists and is finite. The proof relies on the finiteness of the bias v^π , which follows from the invertibility of $I - P^\pi + \mathbf{1}_n(\rho^\pi)^\top$ for an ergodic chain.

4.2.2 (2 Points) Advantage-Like Interpretation

The corresponding state-value function is

$$J^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a)].$$

Show that replacing $Q^\pi(s, a)$ by

$$A^\pi(s, a) = Q^\pi(s, a) - J^\pi(s)$$

does not change the policy gradient.

Answer

Step 1: The average-reward objective. We consider a continuing MDP with finite state/action spaces and a fixed parameterized policy $\pi_\theta(a|s)$. Under ergodicity, the long-run average reward per step is

$$\eta(\theta) = \sum_{s \in \mathcal{S}} \rho^\theta(s) \sum_{a \in \mathcal{A}} \pi_\theta(a|s) R(s, a),$$

where ρ^θ is the unique stationary distribution satisfying $\rho^\theta P^{\pi_\theta} = \rho^\theta$, $\sum_s \rho^\theta(s) = 1$, and $\rho^\theta(s) > 0$ for all s . All terms — ρ^θ , π_θ , and the average reward itself — depend on θ .

Step 2: Policy gradient theorem for average reward (derivation). Define the differential state-value function (bias) $v^\theta(s)$ as the unique solution (up to an additive constant) of the Poisson equation:

$$v^\theta(s) = \sum_a \pi_\theta(a|s) \left[R(s, a) - \eta(\theta) + \sum_{s'} P(s'|s, a) v^\theta(s') \right], \quad \forall s. \quad (1)$$

(Existence and uniqueness up to a constant are guaranteed for an ergodic chain; we fix a particular solution, e.g., by setting $v^\theta(s_0) = 0$ for some reference state.)

Take the gradient of (1) with respect to θ . Using the product rule and noting that $R(s, a)$, $P(s'|s, a)$ are independent of θ :

$$\begin{aligned} \nabla_\theta v^\theta(s) &= \sum_a \nabla_\theta \pi_\theta(a|s) \left[R(s, a) - \eta + \sum_{s'} P(s'|s, a) v^\theta(s') \right] \\ &\quad + \sum_a \pi_\theta(a|s) \left[-\nabla_\theta \eta + \sum_{s'} P(s'|s, a) \nabla_\theta v^\theta(s') \right]. \end{aligned}$$

Now multiply both sides by the stationary probability $\rho^\theta(s)$ and sum over s . We denote $\langle f \rangle = \sum_s \rho^\theta(s) f(s)$ for brevity.

- The left-hand side becomes $\langle \nabla_\theta v^\theta \rangle$.
- The first term on the right gives $\sum_s \rho^\theta(s) \sum_a \nabla_\theta \pi_\theta(a|s) [R(s, a) - \eta + \sum_{s'} P(s'|s, a) v^\theta(s')]$.
- The second term: $-\nabla_\theta \eta \sum_s \rho^\theta(s) \sum_a \pi_\theta(a|s) = -\nabla_\theta \eta \cdot 1$ (since $\sum_a \pi_\theta(a|s) = 1$ and $\sum_s \rho^\theta(s) = 1$).
- The third term (involving $\nabla_\theta v^\theta(s')$) becomes $\sum_{s, s', a} \rho^\theta(s) \pi_\theta(a|s) P(s'|s, a) \nabla_\theta v^\theta(s') = \sum_{s'} (\sum_{s, a} \rho^\theta(s) \pi_\theta(a|s) P(s'|s, a)) \nabla_\theta v^\theta(s')$.

By the stationarity condition, $\sum_{s, a} \rho^\theta(s) \pi_\theta(a|s) P(s'|s, a) = \rho^\theta(s')$. Hence the third term simplifies to $\sum_{s'} \rho^\theta(s') \nabla_\theta v^\theta(s') = \langle \nabla_\theta v^\theta \rangle$.

Now the equation reads

$$\langle \nabla_{\theta} v^{\theta} \rangle = \sum_{s,a} \rho^{\theta}(s) \nabla_{\theta} \pi_{\theta}(a|s) [R(s,a) - \eta + \sum_{s'} P(s'|s,a) v^{\theta}(s')] - \nabla_{\theta} \eta + \langle \nabla_{\theta} v^{\theta} \rangle.$$

Cancel $\langle \nabla_{\theta} v^{\theta} \rangle$ on both sides. Rearranging,

$$\nabla_{\theta} \eta = \sum_s \rho^{\theta}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s) [R(s,a) - \eta + \sum_{s'} P(s'|s,a) v^{\theta}(s')].$$

Step 3: Introduce the differential action-value function Q^{π} . By definition (Section 4.2),

$$Q^{\pi}(s,a) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} (R(s_t, a_t) - \eta) \mid s_0 = s, a_0 = a \right] = R(s,a) - \eta + \sum_{s'} P(s'|s,a) v^{\pi}(s').$$

Thus the bracket in the gradient expression equals exactly $Q^{\pi}(s,a)$. Hence

$$\nabla_{\theta} \eta(\theta) = \sum_s \rho^{\theta}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s) Q^{\pi}(s,a). \quad (2)$$

This is the **policy gradient theorem for the average-reward setting**.

Step 4: Advantage function substitution. Define the differential state-value function

$$J^{\pi}(s) = \mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} [Q^{\pi}(s,a)] = \sum_a \pi_{\theta}(a|s) Q^{\pi}(s,a),$$

and the differential advantage function

$$A^{\pi}(s,a) = Q^{\pi}(s,a) - J^{\pi}(s).$$

Replace $Q^{\pi}(s,a) = A^{\pi}(s,a) + J^{\pi}(s)$ in (2):

$$\begin{aligned} \nabla_{\theta} \eta(\theta) &= \sum_s \rho^{\theta}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s) (A^{\pi}(s,a) + J^{\pi}(s)) \\ &= \sum_s \rho^{\theta}(s) \sum_a \nabla_{\theta} \pi_{\theta}(a|s) A^{\pi}(s,a) + \sum_s \rho^{\theta}(s) J^{\pi}(s) \underbrace{\sum_a \nabla_{\theta} \pi_{\theta}(a|s)}_{=0}. \end{aligned}$$

The inner sum over actions is zero because $\sum_a \pi_{\theta}(a|s) = 1$ for all s , and gradient and sum commute: $\nabla_{\theta} (\sum_a \pi_{\theta}(a|s)) = \nabla_{\theta} 1 = 0$. Therefore the entire second term vanishes.

Step 5: Conclusion. We obtain

$$\boxed{\nabla_{\theta} \eta(\theta) = \sum_{s \in \mathcal{S}} \rho^{\theta}(s) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a|s) A^{\pi}(s,a)}.$$

Replacing the differential action-value $Q^{\pi}(s,a)$ by the advantage $A^{\pi}(s,a) = Q^{\pi}(s,a) - J^{\pi}(s)$ leaves the policy gradient unchanged. This is the average-reward analog of the baseline trick: subtracting any state-dependent function (here $J^{\pi}(s)$) introduces no bias because the score function has zero conditional expectation.

4.3 The Ordinary Policy Gradient

The average-reward policy-gradient theorem gives

$$\nabla \eta(\theta) = \sum_{s,a} \rho^\pi(s) \nabla_\theta \pi_\theta(a|s) Q^\pi(s, a).$$

Using the identity

$$\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s),$$

we can also write this as

$$\nabla \eta(\theta) = \mathbb{E}_{s \sim \rho^\pi, a \sim \pi_\theta(\cdot|s)} [\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s, a)].$$

A standard policy-gradient update is

$$\theta \leftarrow \theta + \alpha \nabla \eta(\theta).$$

This update treats θ -space as Euclidean. In other words, it implicitly measures the size of a parameter step $\Delta\theta$ using

$$\|\Delta\theta\|^2 = \Delta\theta^\top \Delta\theta.$$

The problem is that θ is only a coordinate system. The policy itself is a distribution. Two different parameterizations can represent the same family of policies, but ordinary gradient ascent may behave differently under these two parameterizations.

4.4 The Fisher Geometry of Policies

A parameter vector θ is not important by itself. What matters is the policy distribution $\pi_\theta(\cdot|s)$. If we perturb the parameters from θ to $\theta + \Delta\theta$, the meaningful question is not simply “How large $\Delta\theta$ is” but rather “How much does the action distribution change?”

The Fisher matrix answers this local question. It tells us how sensitive the policy distribution is to small changes in the parameters.

Some directions in parameter space may look large in Euclidean norm but barely change the policy. Other directions may look small but drastically change the action probabilities. The Fisher matrix rescales directions according to their actual effect on the distribution.

For a small perturbation $\Delta\theta$, the quantity

$$\Delta\theta^\top F_s(\theta) \Delta\theta$$

can be interpreted as the local squared distance, at state s , between the old policy distribution $\pi_\theta(\cdot|s)$ and the perturbed policy distribution $\pi_{\theta+\Delta\theta}(\cdot|s)$.

So the Fisher matrix is not measuring distance between parameter vectors directly. It is measuring the local change in the probability distribution represented by those parameters.

The Fisher information matrix for $\pi_\theta(\cdot|s)$ is

$$F_s(\theta) = \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top \right].$$

Since the policy is used across states, the natural metric for the policy class averages this state-wise Fisher information over the stationary distribution:

$$F(\theta) = \mathbb{E}_{s \sim \rho^\pi} [F_s(\theta)].$$

The averaging over $\rho^\pi(s)$ means that changes to the policy are weighted according to how often states are visited under the current policy.

Equivalently,

$$F(\theta) = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top.$$

This matrix measures how much a small parameter perturbation changes the policy distribution. The steepest ascent direction under this metric is

$$\tilde{\nabla}_\eta(\theta) = F(\theta)^{-1} \nabla_\eta(\theta).$$

This is the natural policy gradient.

This optimization problem is the local mathematical version of asking for the steepest ascent direction.

For a small parameter update $\Delta\theta$, we can approximate the change in the objective by the first-order Taylor expansion

$$\eta(\theta + \Delta\theta) \approx \eta(\theta) + \nabla_\eta(\theta)^\top \Delta\theta.$$

Since $\eta(\theta)$ is fixed at the current parameter value, maximizing local improvement is equivalent to maximizing

$$\nabla_\eta(\theta)^\top \Delta\theta.$$

However, without any constraint, this linear expression can be made arbitrarily large just by scaling $\Delta\theta$. Therefore, to compare directions fairly, we fix the size of the update.

If we were using the ordinary Euclidean geometry of parameter space, we would constrain

$$\Delta\theta^\top \Delta\theta = \epsilon.$$

But the natural-gradient viewpoint says that the size of an update should be measured by how much it changes the policy distribution. Locally, this size is measured by the Fisher metric:

$$\Delta\theta^\top F(\theta) \Delta\theta.$$

4.4.1 (4 Points) Natural Gradient as Steepest Ascent Derive the natural-gradient direction by solving the local constrained optimization problem

$$\max_{\Delta\theta} \nabla_\eta(\theta)^\top \Delta\theta$$

subject to

$$\Delta\theta^\top F(\theta) \Delta\theta = \epsilon.$$

Answer

Step 1: Setup. The average-reward objective is $\eta(\theta)$. The ordinary policy gradient is $\nabla_\eta(\theta)$. A small parameter perturbation $\Delta\theta$ changes the policy distribution; the local squared change is measured by the Fisher metric

$$\Delta\theta^\top F(\theta) \Delta\theta, \quad F(\theta) = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top,$$

where ρ^π is the stationary state distribution (ergodicity assumed). We seek the direction that maximises the first-order improvement $\nabla_\eta(\theta)^\top \Delta\theta$ under a fixed policy change $\epsilon > 0$:

$$\max_{\Delta\theta} \nabla_\eta(\theta)^\top \Delta\theta \quad \text{subject to} \quad \Delta\theta^\top F(\theta) \Delta\theta = \epsilon. \quad (1)$$

Step 2: Lagrangian and first-order condition. Form the Lagrangian with multiplier λ (the factor $\frac{1}{2}$ is for convenience):

$$\mathcal{L}(\Delta\theta, \lambda) = \nabla\eta(\theta)^\top \Delta\theta - \frac{\lambda}{2} (\Delta\theta^\top F(\theta) \Delta\theta - \epsilon).$$

Setting $\nabla_{\Delta\theta} \mathcal{L} = 0$ gives

$$\nabla\eta(\theta) - \lambda F(\theta) \Delta\theta = 0 \implies F(\theta) \Delta\theta = \frac{1}{\lambda} \nabla\eta(\theta). \quad (2)$$

Step 3: Invertibility of $F(\theta)$. To solve (2) for $\Delta\theta$ we need $F(\theta)$ to be invertible. We verify this under the assumptions:

Ergodicity $\Rightarrow$ the induced Markov chain is irreducible and aperiodic. For a finite irreducible chain the stationary distribution ρ^π is unique and has full support: $\rho^\pi(s) > 0$ for every $s \in \mathcal{S}$. Each per-state Fisher matrix $F_s(\theta) = \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\nabla \log \pi_\theta \nabla \log \pi_\theta^\top]$ is symmetric positive **semidefinite** (a covariance matrix). Hence the weighted average $F(\theta) = \sum_s \rho^\pi(s) F_s(\theta)$ is also symmetric positive semidefinite. It is **positive definite** (invertible) iff there is no non-zero vector v with $v^\top F_s(\theta) v = 0$ for **all** states s . A direction v satisfies $v^\top F_s(\theta) v = 0$ exactly when the policy distribution $\pi_\theta(\cdot|s)$ does **not** change to first order in direction v at state s .

The policy is assumed **well-parameterised**: the only parameter direction that leaves **all** state-conditional action distributions unchanged is $v = 0$. In other words, the mapping $\theta \mapsto (\pi_\theta(\cdot|s))_{s \in \mathcal{S}}$ is locally injective. Consequently the nullspaces of the $F_s(\theta)$ intersect only at $\{0\}$, and with all weights $\rho^\pi(s) > 0$ the sum $F(\theta)$ is positive definite, hence invertible. (If the assumption fails, $F(\theta)$ may be singular; one then uses the Moore–Penrose pseudoinverse $F(\theta)^\dagger$ instead, which still yields the steepest ascent direction within the effective subspace.)

Step 4: Solve for $\Delta\theta$. With $F(\theta)$ invertible, (2) gives

$$\Delta\theta = \frac{1}{\lambda} F(\theta)^{-1} \nabla\eta(\theta). \quad (3)$$

Substitute (3) into the constraint $\Delta\theta^\top F(\theta) \Delta\theta = \epsilon$:

$$\left(\frac{1}{\lambda} F^{-1} \nabla\eta \right)^\top F \left(\frac{1}{\lambda} F^{-1} \nabla\eta \right) = \frac{1}{\lambda^2} (\nabla\eta)^\top F^{-1} F F^{-1} \nabla\eta = \frac{1}{\lambda^2} (\nabla\eta)^\top F^{-1} \nabla\eta = \epsilon.$$

Thus

$$\lambda = \sqrt{\frac{(\nabla\eta)^\top F^{-1} \nabla\eta}{\epsilon}} (> 0).$$

Step 5: Natural gradient direction. Inserting λ back into (3) yields the optimal parameter perturbation

$$\Delta\theta^* = \sqrt{\frac{\epsilon}{(\nabla\eta)^\top F^{-1} \nabla\eta}} F(\theta)^{-1} \nabla\eta(\theta).$$

The direction of $\Delta\theta^*$ is $F(\theta)^{-1} \nabla\eta(\theta)$, and the scalar prefactor only scales the step size. Therefore the **steepest ascent direction** under the Fisher metric is the **natural policy gradient**

$$\boxed{\tilde{\nabla}\eta(\theta) = F(\theta)^{-1} \nabla\eta(\theta)}.$$

In practice, a natural gradient ascent step uses a learning rate $\alpha > 0$:

$$\theta \leftarrow \theta + \alpha F(\theta)^{-1} \nabla\eta(\theta).$$

4.5 Compatible Function Approximation

Now we connect the natural gradient to a special linear approximation of Q^π .

Define the score feature

$$\psi^\pi(s, a) = \nabla_\theta \log \pi_\theta(a|s).$$

A compatible function approximator has the form

$$f^\pi(s, a; w) = w^\top \psi^\pi(s, a).$$

It is called compatible because it uses exactly the same score features that appear in the policy-gradient expression.

We fit w by minimizing the squared error

$$\epsilon(w, \pi) = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \left(w^\top \psi^\pi(s, a) - Q^\pi(s, a) \right)^2.$$

The important fact is that the best compatible function approximator does not merely approximate Q^π . Its weight vector is exactly the natural-gradient direction.

4.5.1 (3 Points) Compatible Function Approximation and Natural Gradient

Assume w minimizes

$$\epsilon(w, \pi) = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \left(w^\top \psi^\pi(s, a) - Q^\pi(s, a) \right)^2.$$

Show that

$$w = F(\theta)^{-1} \nabla \eta(\theta).$$

Answer

Step 1: Setup. The score feature vector is $\psi^\pi(s, a) = \nabla_\theta \log \pi_\theta(a|s)$. The compatible function approximator is a linear function of these features:

$$f^\pi(s, a; w) = w^\top \psi^\pi(s, a), \quad w \in \mathbb{R}^m.$$

The parameters w are chosen to minimize the weighted squared error between f^π and the true differential action-value function $Q^\pi(s, a)$:

$$\epsilon(w, \pi) = \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \rho^\pi(s) \pi_\theta(a|s) \left(w^\top \psi^\pi(s, a) - Q^\pi(s, a) \right)^2. \quad (1)$$

The weights $\rho^\pi(s) \pi_\theta(a|s)$ are the stationary state-action visitation frequencies under the current policy π_θ .

Step 2: Minimization of the quadratic objective. $\epsilon(w, \pi)$ is a convex quadratic in w . Taking the gradient with respect to w and setting it to zero gives the normal equations:

$$\nabla_w \epsilon(w, \pi) = 2 \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \psi^\pi(s, a) \left(\psi^\pi(s, a)^\top w - Q^\pi(s, a) \right) = 0.$$

Rearranging,

$$\left(\sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \psi^\pi(s, a) \psi^\pi(s, a)^\top \right) w = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \psi^\pi(s, a) Q^\pi(s, a). \quad (2)$$

Step 3: Identify the Fisher matrix and the policy gradient. The matrix on the left-hand side of (2) is exactly the average Fisher information matrix:

$$F(\theta) = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \psi^\pi(s,a) \psi^\pi(s,a)^\top. \quad (3)$$

The vector on the right-hand side is the ordinary policy gradient. By the average-reward policy gradient theorem (derived in the previous subquestion),

$$\nabla_\theta \eta(\theta) = \sum_{s,a} \rho^\pi(s) \nabla_\theta \pi_\theta(a|s) Q^\pi(s,a) = \sum_{s,a} \rho^\pi(s) \pi_\theta(a|s) \psi^\pi(s,a) Q^\pi(s,a). \quad (4)$$

The last equality uses $\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) = \pi_\theta(a|s) \psi^\pi(s,a)$.

Step 4: Solve for w . Substituting (3) and (4) into (2) yields

$$F(\theta) w = \nabla_\theta \eta(\theta).$$

Under the assumption that $F(\theta)$ is invertible (discussed earlier), we obtain

$$w = F(\theta)^{-1} \nabla_\theta \eta(\theta).$$

Thus the weight vector w that best approximates the differential action-value function using the score features is exactly the **natural policy gradient** direction $\tilde{\nabla}_\eta(\theta)$.

Interpretation. This result shows that the natural gradient is not merely a rescaled version of the ordinary gradient; it is the coefficient vector of the optimal linear approximation of Q^π in the basis of score functions. Consequently, a natural gradient step can be implemented by (i) approximating Q^π with a compatible function approximator, and (ii) updating the policy parameters using w as the gradient direction. This insight is the foundation of natural actor-critic algorithms.

4.5.2 (2 Points) Zero-Mean Property

Prove that

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [f^\pi(s, a; w)] = 0.$$

Answer

Step 1: Setup. The compatible function approximator uses the score features $\psi^\pi(s, a) = \nabla_\theta \log \pi_\theta(a|s)$:

$$f^\pi(s, a; w) = w^\top \psi^\pi(s, a), \quad w \in \mathbb{R}^m.$$

For any fixed state s , we consider the expectation over actions drawn from $\pi_\theta(\cdot|s)$.

Step 2: Expectation of the score function. A fundamental property of the score function is that its expected value under the distribution it scores is zero:

$$\begin{aligned}
 \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\psi^\pi(s, a)] &= \sum_{a \in \mathcal{A}} \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s) \\
 &= \sum_{a \in \mathcal{A}} \pi_\theta(a|s) \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} \\
 &= \sum_{a \in \mathcal{A}} \nabla_\theta \pi_\theta(a|s) \\
 &= \nabla_\theta \sum_{a \in \mathcal{A}} \pi_\theta(a|s) \\
 &= \nabla_\theta 1 = 0.
 \end{aligned}$$

The interchange of gradient and sum is valid because the sum is finite.

Step 3: Expectation of the compatible approximator. Now take the expectation of $f^\pi(s, a; w)$:

$$\begin{aligned}
 \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [f^\pi(s, a; w)] &= \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [w^\top \psi^\pi(s, a)] \\
 &= w^\top \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\psi^\pi(s, a)] \\
 &= w^\top \cdot 0 \\
 &= 0.
 \end{aligned}$$

Conclusion. For any weight vector w and any state s ,

$$\mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [f^\pi(s, a; w)] = 0.$$

The compatible function approximator automatically has zero mean under the policy distribution for every state. It can only represent functions that describe the **relative** values of actions – the **advantage**-like component – and cannot capture any state-dependent baseline (which would have non-zero mean). This is why the compatible approximation is naturally advantage-like, and why the optimal weight vector yields exactly the natural policy gradient.

4.6 Natural Gradient and Greedy Policy Improvement

A greedy policy-improvement step using the compatible approximation would choose actions according to

$$a \in \arg \max_{a'} f^\pi(s, a'; w).$$

The surprising connection is that the natural gradient points toward this greedy improvement step.

To see this most clearly, consider an exponential-family policy:

$$\pi_\theta(a|s) = \frac{\exp(\theta^\top \phi_{s,a})}{\sum_b \exp(\theta^\top \phi_{s,b})}.$$

4.6.1 (4 Points) Computing the Score Feature Considering the exponential-family policy, show that the score feature is

$$\psi^\pi(s, a) = \phi_{s,a} - \mathbb{E}_{b \sim \pi_\theta(\cdot|s)} [\phi_{s,b}].$$

Answer

Step 1: Exponential-family policy. We consider the softmax (exponential-family) policy parameterised by $\theta \in \mathbb{R}^m$:

$$\pi_\theta(a|s) = \frac{\exp(\theta^\top \phi_{s,a})}{Z_s(\theta)}, \quad Z_s(\theta) = \sum_{b \in \mathcal{A}} \exp(\theta^\top \phi_{s,b}),$$

where $\phi_{s,a} \in \mathbb{R}^m$ is a fixed feature vector for state-action pair (s, a) , and $Z_s(\theta)$ is the partition function.

Step 2: Log-probability. Taking the logarithm:

$$\log \pi_\theta(a|s) = \theta^\top \phi_{s,a} - \log Z_s(\theta).$$

Step 3: Gradient with respect to θ . The score feature is defined as $\psi^\pi(s, a) = \nabla_\theta \log \pi_\theta(a|s)$. Differentiating term by term:

- $\nabla_\theta (\theta^\top \phi_{s,a}) = \phi_{s,a}$ (since $\phi_{s,a}$ does not depend on θ).
- $\nabla_\theta \log Z_s(\theta)$ requires the chain rule.

Step 4: Gradient of the log-partition function.

$$\begin{aligned} \nabla_\theta \log Z_s(\theta) &= \frac{1}{Z_s(\theta)} \nabla_\theta Z_s(\theta) \\ &= \frac{1}{Z_s(\theta)} \nabla_\theta \sum_{b \in \mathcal{A}} e^{\theta^\top \phi_{s,b}} \\ &= \frac{1}{Z_s(\theta)} \sum_{b \in \mathcal{A}} e^{\theta^\top \phi_{s,b}} \phi_{s,b} \\ &= \sum_{b \in \mathcal{A}} \frac{e^{\theta^\top \phi_{s,b}}}{Z_s(\theta)} \phi_{s,b} \\ &= \sum_{b \in \mathcal{A}} \pi_\theta(b|s) \phi_{s,b} \\ &= \mathbb{E}_{b \sim \pi_\theta(\cdot|s)} [\phi_{s,b}]. \end{aligned}$$

Thus the gradient of the log-partition function is exactly the expected feature vector under the current policy.

Step 5: Assemble the score feature. Putting the two parts together:

$$\begin{aligned} \psi^\pi(s, a) &= \nabla_\theta \log \pi_\theta(a|s) \\ &= \phi_{s,a} - \nabla_\theta \log Z_s(\theta) \\ &= \phi_{s,a} - \mathbb{E}_{b \sim \pi_\theta(\cdot|s)} [\phi_{s,b}]. \end{aligned}$$

Conclusion. For the exponential-family policy, the score feature is the **centered** feature vector:

$$\boxed{\psi^\pi(s, a) = \phi_{s,a} - \mathbb{E}_{b \sim \pi_\theta(\cdot|s)} [\phi_{s,b}]}.$$

This expression subtracts the state-conditional mean of the features, so $\mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\psi^\pi(s, a)] = 0$, consistent with the zero-mean property of compatible function approximators.

4.6.2 (4 Points) Large-Step Greedy Limit Define

$$\pi_\infty(a|s) = \lim_{\alpha \rightarrow \infty} \pi_{\theta + \alpha \tilde{\nabla} \eta(\theta)}(a|s).$$

Show that

$$\pi_\infty(a|s) \neq 0$$

if and only if

$$a \in \arg \max_{a'} f^\pi(s, a'; w).$$

Answer

Step 1: Setup. From the compatible function approximation result, the natural gradient direction is exactly the weight vector of the optimal compatible function approximator:

$$\tilde{\nabla} \eta(\theta) = F(\theta)^{-1} \nabla \eta(\theta) = w.$$

We consider an exponential-family policy

$$\pi_\theta(a|s) = \frac{\exp(\theta^\top \phi_{s,a})}{\sum_b \exp(\theta^\top \phi_{s,b})}.$$

The updated parameters after a natural gradient step of size $\alpha > 0$ are

$$\theta' = \theta + \alpha \tilde{\nabla} \eta(\theta) = \theta + \alpha w.$$

We examine the limiting policy as $\alpha \rightarrow \infty$:

$$\pi_\infty(a|s) = \lim_{\alpha \rightarrow \infty} \pi_{\theta + \alpha w}(a|s).$$

Step 2: Expand the updated policy.

$$\pi_{\theta + \alpha w}(a|s) = \frac{\exp((\theta + \alpha w)^\top \phi_{s,a})}{\sum_b \exp((\theta + \alpha w)^\top \phi_{s,b})} = \frac{\exp(\theta^\top \phi_{s,a} + \alpha w^\top \phi_{s,a})}{\sum_b \exp(\theta^\top \phi_{s,b} + \alpha w^\top \phi_{s,b})}.$$

Step 3: Relate $w^\top \phi_{s,a}$ to the compatible approximation. From the previous subquestion, the score feature for the exponential family is $\psi^\pi(s, a) = \phi_{s,a} - \mathbb{E}_{b \sim \pi_\theta(\cdot|s)}[\phi_{s,b}]$. The compatible function approximator is $f^\pi(s, a; w) = w^\top \psi^\pi(s, a)$. Hence

$$f^\pi(s, a; w) = w^\top \phi_{s,a} - w^\top \mathbb{E}_{b \sim \pi_\theta(\cdot|s)}[\phi_{s,b}].$$

Define the state-dependent constant (independent of a)

$$C(s) = w^\top \mathbb{E}_{b \sim \pi_\theta(\cdot|s)}[\phi_{s,b}].$$

Then

$$w^\top \phi_{s,a} = f^\pi(s, a; w) + C(s).$$

Step 4: Substitute and simplify. Insert this into the policy expression:

$$\begin{aligned} \pi_{\theta + \alpha w}(a|s) &= \frac{\exp(\theta^\top \phi_{s,a} + \alpha f^\pi(s, a; w) + \alpha C(s))}{\sum_b \exp(\theta^\top \phi_{s,b} + \alpha f^\pi(s, b; w) + \alpha C(s))} \\ &= \frac{e^{\alpha C(s)} \exp(\theta^\top \phi_{s,a} + \alpha f^\pi(s, a; w))}{e^{\alpha C(s)} \sum_b \exp(\theta^\top \phi_{s,b} + \alpha f^\pi(s, b; w))} \\ &= \frac{\exp(\theta^\top \phi_{s,a} + \alpha f^\pi(s, a; w))}{\sum_b \exp(\theta^\top \phi_{s,b} + \alpha f^\pi(s, b; w))}. \end{aligned}$$

The $e^{\alpha C(s)}$ factors cancel; the state-dependent offset disappears.

Step 5: Take the limit $\alpha \rightarrow \infty$. Let $M(s) = \max_{a' \in \mathcal{A}} f^\pi(s, a'; w)$. Factor $e^{\alpha M(s)}$ from the numerator and denominator:

$$\pi_{\theta+\alpha w}(a|s) = \frac{e^{\alpha M(s)} \exp(\theta^\top \phi_{s,a} + \alpha (f^\pi(s, a; w) - M(s)))}{e^{\alpha M(s)} \sum_b \exp(\theta^\top \phi_{s,b} + \alpha (f^\pi(s, b; w) - M(s)))} = \frac{\exp(\theta^\top \phi_{s,a} + \alpha \Delta_a(s))}{\sum_b \exp(\theta^\top \phi_{s,b} + \alpha \Delta_b(s))},$$

where $\Delta_a(s) = f^\pi(s, a; w) - M(s) \leq 0$.

- If $a \notin \arg \max_{a'} f^\pi(s, a'; w)$, then $\Delta_a(s) < 0$. As $\alpha \rightarrow \infty$, $\alpha \Delta_a(s) \rightarrow -\infty$, so the numerator tends to 0, while the denominator remains at least $\sum_{b: \Delta_b(s)=0} \exp(\theta^\top \phi_{s,b}) > 0$. Hence $\pi_\infty(a|s) = 0$.
- If $a \in \arg \max_{a'} f^\pi(s, a'; w)$, then $\Delta_a(s) = 0$. The numerator is $\exp(\theta^\top \phi_{s,a})$, finite and positive. All suboptimal terms in the denominator vanish in the limit, leaving only the sum over optimal actions. Therefore

$$\pi_\infty(a|s) = \frac{\exp(\theta^\top \phi_{s,a})}{\sum_{b: f^\pi(s, b; w) = M(s)} \exp(\theta^\top \phi_{s,b})} > 0.$$

Conclusion. $\pi_\infty(a|s) \neq 0$ precisely when a is an action that maximises the compatible function approximator $f^\pi(s, a; w)$. In other words,

$$\pi_\infty(a|s) > 0 \iff a \in \arg \max_{a'} f^\pi(s, a'; w).$$

A very large natural-gradient step thus implements a greedy policy improvement: the updated policy concentrates its probability mass on the action(s) with the highest estimated advantage according to the compatible approximation. This deep connection explains why the natural gradient is not merely a rescaled version of the ordinary gradient—it points directly toward the greedy action choices implied by the current value-function estimate.

4.7 Local Policy Improvement for General Policies

The previous subquestion used exponential-family policies and a large step size. For a general smooth policy class, we can still obtain a local statement.

Let

$$\theta' = \theta + \alpha \tilde{\nabla} \eta(\theta).$$

For a general smooth policy, the infinite-step argument no longer applies. However, for small α , the natural-gradient update still has a local policy-improvement interpretation: it changes action probabilities according to the compatible approximation $f^\pi(s, a; w)$.

4.7.1 (4 Points) Local Natural-Gradient Update For a general smooth policy, prove the local first-order relationship between $\pi_{\theta'}(a|s)$ and $\pi_\theta(a|s)$ under the update $\theta' = \theta + \alpha \tilde{\nabla} \eta(\theta)$. Then, show that for sufficiently small α , to first order, actions with larger values of $f^\pi(s, a; w)$ receive larger multiplicative increases in probability.

Answer

Step 1: Setup and Taylor expansion. Let π_θ be a general smooth policy (continuously differentiable in θ). The natural gradient update is $\theta' = \theta + \alpha \tilde{\nabla} \eta(\theta)$ with $\tilde{\nabla} \eta(\theta) = w$. We consider $\Delta\theta = \theta' - \theta = \alpha w$, and expand $\pi_{\theta'}(a|s)$ around θ to first order:

$$\pi_{\theta'}(a|s) = \pi_\theta(a|s) + \nabla_\theta \pi_\theta(a|s)^\top \Delta\theta + \mathcal{R}(\Delta\theta),$$

where the remainder $\mathcal{R}(\Delta\theta)$ satisfies $\mathcal{R}(\Delta\theta) = O(\|\Delta\theta\|^2)$. Because w is fixed and $\|\Delta\theta\|^2 = \alpha^2 \|w\|^2$, we have $\mathcal{R}(\Delta\theta) = O(\alpha^2)$ as $\alpha \rightarrow 0$.

Step 2: Express $\nabla_\theta \pi_\theta$ via the score function. Using the identity $\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)$, we obtain

$$\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s) \psi^\pi(s, a),$$

where $\psi^\pi(s, a) = \nabla_\theta \log \pi_\theta(a|s)$ is the score feature.

Step 3: Substitute $\Delta\theta$ and the score form.

$$\begin{aligned} \pi_{\theta'}(a|s) &= \pi_\theta(a|s) + \pi_\theta(a|s) \psi^\pi(s, a)^\top (\alpha w) + O(\alpha^2) \\ &= \pi_\theta(a|s) \left[1 + \alpha w^\top \psi^\pi(s, a) \right] + O(\alpha^2). \end{aligned}$$

Step 4: Identify the compatible function approximator. By definition, the compatible approximator with weight vector w is

$$f^\pi(s, a; w) = w^\top \psi^\pi(s, a).$$

Hence

$$\pi_{\theta'}(a|s) = \pi_\theta(a|s) \left[1 + \alpha f^\pi(s, a; w) \right] + O(\alpha^2). \quad (1)$$

Step 5: Interpretation – multiplicative probability changes. Equation (1) shows that, to first order in α , the updated probability is the old probability multiplied by $1 + \alpha f^\pi(s, a; w)$:

- If $f^\pi(s, a; w) > 0$ (action a is estimated to be better than the state-average), then $\pi_{\theta'}(a|s) > \pi_\theta(a|s)$ for sufficiently small α .
- If $f^\pi(s, a; w) < 0$, then $\pi_{\theta'}(a|s) < \pi_\theta(a|s)$.

Thus actions with larger f^π receive larger multiplicative increases in probability. Moreover, from the zero-mean property, $\mathbb{E}_{a \sim \pi_\theta} [f^\pi(s, a; w)] = 0$, which guarantees $\sum_a \pi_{\theta'}(a|s) = 1 + \alpha \cdot 0 + O(\alpha^2) = 1 + O(\alpha^2)$, consistent with normalization up to first order.

Conclusion. For any smooth policy and a small natural-gradient step, the local update rule is a multiplicative modulation of the action probabilities by $1 + \alpha f^\pi(s, a; w)$. Therefore, the natural gradient direction w locally pushes the policy toward actions with higher compatible advantage f^π , implementing a soft, local version of greedy policy improvement.

4.8 Why This is Not Newton's Method

The Fisher matrix is not generally the Hessian of the reinforcement-learning objective. The Fisher matrix is

$$F(\theta) = \mathbb{E}_{s,a} \left[\nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top \right].$$

It measures curvature of the policy distribution.

4.8.1 (2 Points) Conceptual Comparison Explain why the natural gradient is not the same as Newton's method. What does the Fisher matrix measure, and what additional quantities appear in the Hessian of the RL objective (write the additional terms)?

Answer

The natural gradient is **not** Newton's method. Newton's method uses the inverse of the Hessian $H(\theta) = \nabla^2 \eta(\theta)$ of the average-reward objective, which measures the curvature of the return function itself. The Fisher matrix $F(\theta)$ only measures the local curvature of the **policy distribution** – specifically, it captures how much a small parameter perturbation changes the action probabilities, via the expected outer product of the score functions. It contains no information about the reward function, the transition dynamics, or the value function.

The Hessian of the RL objective contains additional terms beyond $F(\theta)$. By differentiating the policy gradient theorem, one finds

$$H(\theta) = F(\theta) + \mathbb{E}_{s,a} \left[\nabla_{\theta} \log \pi_{\theta} (\nabla_{\theta} Q^{\pi})^{\top} \right] + \text{terms involving } \nabla_{\theta} \rho^{\pi},$$

where Q^{π} is the differential action-value function and ρ^{π} is the stationary state distribution. The extra terms arise because Q^{π} and ρ^{π} themselves depend on the policy parameters θ . The natural gradient ignores these second-order effects and simply rescales the ordinary gradient according to the geometry of the policy manifold, making it invariant to reparameterization. In contrast, Newton's method uses the full objective curvature to achieve quadratic local convergence, but is not reparameterization-invariant.

4.9 Natural Gradient as a Small Policy-Distribution Step

The previous sections showed that the Fisher matrix defines a geometry on policy space. We now make this more precise using KL divergence.

A parameter update can be small in Euclidean norm but still cause a large change in the policy distribution. Conversely, a parameter update can look large in Euclidean norm but barely change the action probabilities. Therefore, the natural question is not only "how large is $\Delta\theta$?" but rather "How much does the policy distribution change?"

For a fixed state s , consider the KL divergence between the old policy and the new policy:

$$D_{\text{KL}}(\pi_{\theta}(\cdot|s) \parallel \pi_{\theta+\Delta\theta}(\cdot|s)).$$

Averaging over states gives the local policy-distribution change $\bar{D}_{\text{KL}}(\theta, \theta + \Delta\theta)$.

4.9.1 (2 Points) KL Divergence and the Fisher Matrix Write the Taylor expansion of $\bar{D}_{\text{KL}}(\theta, \theta + \Delta\theta)$, around $\Delta\theta = 0$ up to second order. Then explain why the natural-gradient direction is the steepest ascent direction under a local KL constraint.

Thus the Fisher matrix is the local quadratic approximation to KL divergence between nearby policies.

This explains the difference between ordinary gradient ascent and natural gradient ascent. Ordinary gradient ascent chooses the direction that gives the largest first-order improvement for a fixed Euclidean parameter step:

$$\Delta\theta^{\top} \Delta\theta = \epsilon.$$

Natural gradient ascent instead chooses the direction that gives the largest first-order improvement for a fixed local policy-distribution step:

$$\frac{1}{2} \Delta \theta^\top F(\theta) \Delta \theta = \epsilon.$$

So the natural gradient is trying to keep the actual change in the policy distribution small.

Ordinary PG controls parameter movement. Natural PG controls policy-distribution movement.

Answer

Step 1: KL divergence for a fixed state. For a fixed state s , the Kullback–Leibler divergence between the original policy $\pi_\theta(\cdot|s)$ and a perturbed policy $\pi_{\theta+\Delta\theta}(\cdot|s)$ is

$$D_{\text{KL}}(\pi_\theta(\cdot|s) \parallel \pi_{\theta+\Delta\theta}(\cdot|s)) = \sum_{a \in \mathcal{A}} \pi_\theta(a|s) \log \frac{\pi_\theta(a|s)}{\pi_{\theta+\Delta\theta}(a|s)}.$$

We want its Taylor expansion in $\Delta\theta$ around $\mathbf{0}$ up to second order.

Step 2: Expand the log-ratio. Write

$$\begin{aligned} \log \frac{\pi_\theta(a|s)}{\pi_{\theta+\Delta\theta}(a|s)} &= \log \pi_\theta(a|s) - \log \pi_{\theta+\Delta\theta}(a|s) \\ &= -[\log \pi_{\theta+\Delta\theta}(a|s) - \log \pi_\theta(a|s)]. \end{aligned}$$

Expand $\log \pi_{\theta+\Delta\theta}(a|s)$ around θ :

$$\log \pi_{\theta+\Delta\theta}(a|s) = \log \pi_\theta(a|s) + \nabla_\theta \log \pi_\theta(a|s)^\top \Delta\theta + \frac{1}{2} \Delta\theta^\top \nabla_\theta^2 \log \pi_\theta(a|s) \Delta\theta + O(\|\Delta\theta\|^3).$$

Hence

$$-[\log \pi_{\theta+\Delta\theta}(a|s) - \log \pi_\theta(a|s)] = -\nabla_\theta \log \pi_\theta(a|s)^\top \Delta\theta - \frac{1}{2} \Delta\theta^\top \nabla_\theta^2 \log \pi_\theta(a|s) \Delta\theta + O(\|\Delta\theta\|^3).$$

Step 3: Insert into the KL expression.

$$\begin{aligned} D_{\text{KL}} &= \sum_a \pi_\theta(a|s) \left[-\nabla_\theta \log \pi_\theta(a|s)^\top \Delta\theta - \frac{1}{2} \Delta\theta^\top \nabla_\theta^2 \log \pi_\theta(a|s) \Delta\theta \right] + O(\|\Delta\theta\|^3) \\ &= \underbrace{-\mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s)]^\top \Delta\theta}_{=0} - \frac{1}{2} \Delta\theta^\top \underbrace{\mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta^2 \log \pi_\theta(a|s)] \Delta\theta}_{=-F_s(\theta) \Delta\theta} + O(\|\Delta\theta\|^3). \end{aligned}$$

The first term vanishes because $\mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s)] = \nabla_\theta \sum_a \pi_\theta(a|s) = \nabla_\theta 1 = 0$. For the second term, we use the identity (derived below) that $\mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta^2 \log \pi_\theta(a|s)] = -F_s(\theta)$, where $F_s(\theta) = \mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta \nabla_\theta \log \pi_\theta^\top]$ is the per-state Fisher matrix.

Thus

$$D_{\text{KL}}(\pi_\theta(\cdot|s) \parallel \pi_{\theta+\Delta\theta}(\cdot|s)) = \frac{1}{2} \Delta\theta^\top F_s(\theta) \Delta\theta + O(\|\Delta\theta\|^3).$$

Step 4: Proof of the identity $\mathbb{E}[\nabla_\theta^2 \log \pi_\theta] = -F_s(\theta)$. Differentiate the score function inside

the expectation: $\nabla_{\theta}^2 \log \pi_{\theta} = \nabla_{\theta} \left(\frac{\nabla_{\theta} \pi_{\theta}}{\pi_{\theta}} \right) = \frac{\nabla_{\theta}^2 \pi_{\theta}}{\pi_{\theta}} - \frac{\nabla_{\theta} \pi_{\theta} \nabla_{\theta} \pi_{\theta}^{\top}}{\pi_{\theta}^2}$. Taking expectation,

$$\begin{aligned} \mathbb{E}_a[\nabla_{\theta}^2 \log \pi_{\theta}] &= \sum_a \pi_{\theta}(a|s) \frac{\nabla_{\theta}^2 \pi_{\theta}(a|s)}{\pi_{\theta}(a|s)} - \sum_a \pi_{\theta}(a|s) \frac{\nabla_{\theta} \pi_{\theta}(a|s) \nabla_{\theta} \pi_{\theta}(a|s)^{\top}}{\pi_{\theta}(a|s)^2} \\ &= \nabla_{\theta}^2 \sum_a \pi_{\theta}(a|s) - \mathbb{E}_a \left[\frac{\nabla_{\theta} \pi_{\theta}}{\pi_{\theta}} \frac{\nabla_{\theta} \pi_{\theta}^{\top}}{\pi_{\theta}} \right] \\ &= 0 - \mathbb{E}_a[\nabla_{\theta} \log \pi_{\theta} \nabla_{\theta} \log \pi_{\theta}^{\top}] \\ &= -F_s(\theta). \end{aligned}$$

Step 5: Average over the stationary distribution. Define the average KL divergence

$$\hat{D}_{\text{KL}}(\theta, \theta + \Delta\theta) = \sum_{s \in \mathcal{S}} \rho^{\pi}(s) D_{\text{KL}}(\pi_{\theta}(\cdot|s) \parallel \pi_{\theta+\Delta\theta}(\cdot|s)).$$

Substituting the expansion for each state,

$$\begin{aligned} \hat{D}_{\text{KL}} &= \sum_s \rho^{\pi}(s) \left(\frac{1}{2} \Delta\theta^{\top} F_s(\theta) \Delta\theta + O(\|\Delta\theta\|^3) \right) \\ &= \frac{1}{2} \Delta\theta^{\top} \left(\sum_s \rho^{\pi}(s) F_s(\theta) \right) \Delta\theta + O(\|\Delta\theta\|^3) \\ &= \frac{1}{2} \Delta\theta^{\top} F(\theta) \Delta\theta + O(\|\Delta\theta\|^3), \end{aligned}$$

where $F(\theta) = \sum_s \rho^{\pi}(s) F_s(\theta)$ is the average Fisher matrix.

Thus **the Fisher matrix is the local quadratic approximation to the average KL divergence.**

Step 6: Steepest ascent under a KL constraint. We now seek the parameter update $\Delta\theta$ that maximises the first-order improvement of the average-reward objective

$$\eta(\theta + \Delta\theta) \approx \eta(\theta) + \nabla \eta(\theta)^{\top} \Delta\theta,$$

while keeping the policy distribution change small, i.e. $\hat{D}_{\text{KL}} \leq \epsilon$. To first order in $\Delta\theta$, this constraint is $\frac{1}{2} \Delta\theta^{\top} F(\theta) \Delta\theta \leq \epsilon$. To find the direction, we fix the constraint as equality (the maximum will be on the boundary). This is exactly the constrained optimisation

$$\max_{\Delta\theta} \nabla \eta(\theta)^{\top} \Delta\theta \quad \text{subject to} \quad \frac{1}{2} \Delta\theta^{\top} F(\theta) \Delta\theta = \epsilon.$$

Solving with a Lagrange multiplier (as in the earlier subquestion) yields the natural gradient direction

$$\Delta\theta \propto F(\theta)^{-1} \nabla \eta(\theta),$$

i.e. $\tilde{\nabla} \eta(\theta) = F(\theta)^{-1} \nabla \eta(\theta)$.

Thus, while ordinary gradient ascent takes the steepest step for a fixed Euclidean parameter change $\Delta\theta^{\top} \Delta\theta = \epsilon$, natural gradient ascent takes the steepest step for a fixed policy-distribution change measured by KL divergence. The Fisher matrix converts the geometric cost of parameter movement into the actual change in the policy probabilities.

4.10 MiniTetris

The previous subsections developed the natural policy gradient from a theoretical point of view. To make these ideas more concrete, you will now implement the accompanying notebook. The goal of the notebook is not only to train a policy, but also to visualize the geometry behind natural policy gradient.

The notebook implements a small Tetris-like control problem in which the policy is a softmax over afterstate features:

$$\pi_{\theta}(a|s) = \frac{\exp(\theta^{\top} \phi_{s,a})}{\sum_b \exp(\theta^{\top} \phi_{s,b})}.$$

Your implementation should help you understand the following ideas more clearly:

- how the score feature $\nabla_{\theta} \log \pi_{\theta}(a|s)$ is computed for a softmax policy;
- how the policy-gradient vector g and the Fisher matrix F can be estimated from rollouts;
- how the natural-gradient direction $d_{\text{NPG}} = (F + \lambda I)^{-1}g$ differs from the ordinary policy-gradient direction $d_{\text{PG}} = g$;
- why comparing PG and NPG using the same Euclidean step size is different from comparing them using the same local KL step size;
- how the compatible function approximation explains the greedy-action behavior of large natural-gradient updates.

4.10.1 (15 Points) Implementation 2: PG and NPG in MiniTetris Implement the described policy gradient and natural policy gradient methods in the provided notebook.

Answer

4.10.2 (1 Point) PG vs. NPG In the notebook we compare ordinary policy gradient and natural policy gradient under both fixed-learning-rate updates and KL-scaled updates. Explain whether the observed behavior agrees with the theoretical interpretation developed in this section.

Answer